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DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization

A Capstone Project

Presented to the Dean of the School of Communication,

Multimedia and Computer Studies

St. Dominic College of Asia

Bacoor, Cavite

In partial fulfillment

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Bachelor of Science in Information Technology

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Technology continues to reshape how institutions and organizations carry out everyday administrative work. Processes that once relied on paper and manual effort are steadily being replaced by digital tools that speed up operations, cut down on errors, and make information easier to access. Within education specifically, this shift is visible in areas such as student records management, grading, and attendance tracking. As digital adoption grows, schools are increasingly expected to move away from repetitive, manual routines that are vulnerable to mistakes and inefficiency (Santos et al., 2021).

Among these administrative routines, attendance monitoring holds particular importance for schools, seminars, and training programs. It gives organizers a way to confirm who took part in an activity and whether participants fulfilled the requirements expected of them. In academic settings, attendance often factors into how student participation and performance are assessed, since consistent presence is closely tied to how much a student engages with and benefits from instruction (Dassanayake & Wanniarachchi, 2021). Because of this link between attendance and meaningful participation, keeping accurate records is essential to preserving fairness in how seminars, lectures, and similar activities are evaluated.

Even so, many institutions still depend on traditional, paper-based methods for recording attendance, typically through sign-in sheets where participants write and sign their names. While straightforward, this approach creates several downstream problems. It tends to slow down large seminars, where dozens or hundreds of participants must sign in before an

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activity can proceed, cutting into time that could otherwise go toward the actual event. The sheets also usually need to be manually transcribed into a computer system afterward to produce usable reports, adding yet another layer of work for administrators (Santos et al., 2021).

Manual recording also opens the door to human error and dishonesty. Because the process depends on handwriting and manual transcription, mistakes are common names may be misspelled, entries misread, or records mismatched during encoding. Beyond simple error, paper sign-in sheets are also vulnerable to proxy attendance, where someone signs on behalf of an absent participant. This undermines the credibility of attendance data and can result in people receiving credit for events they never actually attended (Dassanayake & Wanniarachchi, 2021).

Findings from Dassanayake and Wanniarachchi (2021) reinforce these concerns, noting that many institutions still require students to sign in by hand at every session, a practice that consumes class time and can delay the start of activities. Their study also linked manual sign-in processes to inaccurate records and attendance manipulation, both of which weaken confidence in the data collected. These findings are part of why researchers increasingly call for automated alternatives that can record attendance more quickly and more reliably.

In response, a growing number of institutions have turned to automated attendance systems. These systems record participation electronically, allowing administrators to collect, store, and review attendance data with far less manual effort. Automation also shortens the

time needed to generate reports and reduces the errors that come with manual encoding, while supporting a shift toward paperless recordkeeping (Santos et al., 2021).

One of the more established technologies behind this shift is Radio Frequency Identification (RFID), which allows a participant's identity to be verified simply by scanning a card or tag. Each scan is logged with a timestamp, giving administrators an accurate record of when a participant checked in. RFID's speed and reliability have made it common not just in attendance tracking but in access control and inventory management as well, largely because it removes the need for participants to manually record their own information (Godase et al., 2023).

RFID also strengthens the security of attendance records. Since each card is uniquely tied to one individual, it becomes much harder for someone to falsify attendance on another person's behalf, and centralized storage of scan data allows administrators to review participation history whenever needed. These characteristics make RFID a practical fit for improving attendance monitoring in both academic and professional contexts.

However, attendance monitoring covers only part of the administrative workflow. Issuing certificates to participants afterward is an equally important, and often equally manual, process. While certificates typically serve as proof of participation for academic or professional purposes, manual generation and distribution introduce operational delays and human error. Organizers usually have to cross-check attendance records by hand before certificates can be released, a task that becomes increasingly time-consuming as the number of participants grows .

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This manual verification step also raises the risk that certificates may end up issued to individuals who didn't actually meet the attendance requirement, which damages the credibility of the certification process as a whole. As a response, many institutions have started adopting electronic certificates that can be generated and distributed digitally, an approach that speeds up delivery, reduces printing costs, and supports more sustainable, paperless operations (Porwal & Rastogi, 2020).

Still, even automated attendance systems tend to stop short of addressing what actually happens once large numbers of participants are physically present at an event. Many systems can log an entry accurately but do not verify whether a participant remained for the full required duration, since only the initial scan is typically captured leaving organizers with an accurate check-in time but no confirmation of genuine, complete attendance. This gap becomes more pronounced as event size grows: even digital and biometric attendance systems have been found to experience processing delays and reduced reliability in large-scale environments (Bugingo et al., 2025), and digital identification alone does not eliminate misuse, since identification methods can still be exploited even once manual sign-in sheets are removed (Ishaq & Bibi, 2023; Vadwala, 2024). Beyond individual-level validation, organizers are also left with little visibility into what is unfolding at the event level in real time there is typically no way to tell whether a venue is nearing capacity, whether entry points are creating congestion, or whether turnout is likely to exceed what a space can reasonably accommodate, until the problem has already surfaced. These are the specific gaps the proposed system is designed to close. Beyond validating attendance duration, DC Space is designed to incorporate a Gemini AI layer intended to support administrators with analytics

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such as participation trends, engagement summaries, and as the platform matures predictive insights like congestion patterns during high-traffic events, giving administrators a clearer picture of how facilities and resources are actually being used.

Given these persistent gaps, there is a clear need for a system that unifies RFID-based attendance logging, digital record-keeping, and intelligent validation into a single, cohesive workflow. Such a system would also benefit from working alongside related institutional tools, for instance, DC Space is designed to eventually integrate with iRoomReserve, the school's room reservation platform, so that event organizers can link an approved venue directly during event creation rather than coordinating room bookings separately.

It is within this context that the study proposes the development of DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. The system is designed around three main user roles students and faculty who tap in and out using RFID-enabled IDs, organization officers who submit and manage events, and administrators who oversee approvals, monitor attendance, and manage certificate issuance all accessible through a web platform, with a companion mobile experience planned as the system matures.

Beyond replacing manual sign-in sheets, DC Space aims to ensure that only registered, verified participants can log attendance, that attendance duration is validated before any certificate is issued, and that certificates are generated and delivered automatically once requirements are met, reducing the administrative workload placed on event organizers while making the process more transparent for participants.

Ultimately, by bringing together RFID technology, AI-assisted validation and analytics, and automated certificate issuance, DC Space is intended to make attendance monitoring and certificate distribution at St. Dominic College of Asia is faster, more accurate, and more accountable while laying the groundwork for more advanced, data-driven event planning in the future.

1.2 Problem Statement

At St. Dominic College of Asia, the handling of student attendance and certificate issuance remains largely manual and disconnected across different processes. While the institution has adopted some basic digital tools for event registration, the actual work of tracking attendance and preparing certificates still depends heavily on paper sign-in sheets and manual documentation. This reliance on manual handling leaves the process open to errors, delays, and duplicated effort, all of which chip away at the accuracy and efficiency of institutional record-keeping.

This pattern is consistent with broader findings in the literature. Nkata (2024) points out that traditional attendance recording tends to be unreliable precisely because it depends on manual effort; it is slow, prone to human error, and difficult to use as a trustworthy basis for verifying who actually participated. Salunkhe et al. (2025) reach a similar conclusion, observing that manual attendance systems in schools are time-consuming and error-prone, and arguing that automated alternatives such as RFID-based tracking are needed to make record-keeping more dependable. Boddupalli et al. (2024) extend this concern to certificate handling, noting that manual certificate management tends to create administrative bottlenecks, slow down distribution, and increase the likelihood of inconsistent records

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underscoring the need for automated certificate generation. Taken together, these findings point to a common thread: the absence of a single, real-time system capable of reliably tracking participation, confirming attendance, and issuing certificates without manual intervention at each step.

Communication gaps compound the problem. Usiobaifo and Roseline (2025) emphasize that timely reminders and notifications play a meaningful role in keeping students aware of upcoming academic and extracurricular activities; without them, students risk missing events simply because they weren't properly informed. This issue is not just theoretical in the context of DC Space: even where a notification feature exists, it is currently built to generate messages on the user's own device by piecing together their event and attendance data locally, rather than being pushed centrally by the system. This means updates are only as current as the last time a student opened the app, which limits how reliably reminders reach students when they're actually needed.

A related gap exists in how student feedback is currently collected. A feedback submission feature is already in place, but it does not yet save responses to the system's records meaning that whatever input students provide is effectively lost after submission rather than being available for administrators to review or act on. This disconnects the feedback loop that could otherwise help improve future event planning.

There is also a security dimension to manual and semi-automated attendance tracking that deserves attention: without a mechanism to confirm that the person tapping in is actually the account holder, RFID-based systems remain just as susceptible to proxy attendance as

paper sign-in sheets, only faster. Any automated solution needs to close this gap directly rather than simply digitizing the same vulnerability.

To address these overlapping issues, the proposed system introduces an RFID-integrated platform for managing student attendance and certificate distribution across academic and extracurricular activities. The system ties each RFID scan to the signed-in account it belongs to, allowing administrators to verify participation instantly while ensuring that attendance cannot be logged on someone else's behalf. This removes the need for manual sign-in sheets and paper-based certificate handling, cutting down on both administrative workload and the errors that come with manual processes.

Beyond attendance, the system automates certificate generation once attendance requirements are met, and is designed to deliver timely updates to students regarding their participation status and certificate eligibility addressing the reliability gap left by the current client-side notification approach. A structured feedback pathway is also part of the system's intended scope, ensuring that student input is properly captured and made available to administrators rather than disappearing after submission. Through centralized data management and end-to-end automation, the system is designed to improve the accuracy, efficiency, and transparency of attendance and certificate handling creating a more dependable experience for students, faculty, and administrators alike.

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1.3 Objectives of the Study

This study aimed to design and develop DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization to improve event planning, attendance monitoring, crowd management, participant engagement, and post-event evaluation for academic and extracurricular activities at St. Dominic College of Asia.

Specifically, it aimed to:

- Develop an event management and participant engagement system that allowed students and faculty to browse and register for events, submit required files, receive notifications, answer mid-event and post-event surveys, and upload event photos when permitted. It also allowed event organizers to create and manage events, handle participant registration, review submitted requirements, and manage event activities throughout the event lifecycle.
- Implement an RFID-based attendance and security monitoring system that allowed students and faculty to record their attendance by tapping their RFID-enabled school ID cards while providing real-time attendance logs and attendance validation. The system also detected duplicate entry attempts, suspicious RFID scanning behavior, and invalid scans to improve the reliability of attendance records.
- Develop an AI-powered event intelligence module that predicted expected attendee counts, analyzed attendance flow, monitored crowd density, identified possible congestion, and provided event insights to help organizers and administrators make better decisions before and during events.
- Develop automated certificate and event report management that generated digital certificates based on validated attendance records and produced AI-assisted event

reports containing attendance results, participant feedback, event outcomes, recommendations, and other relevant event information.

- Evaluate the developed system in terms of functionality, usability, efficiency, reliability, accuracy, and user satisfaction to determine its effectiveness in improving event management, attendance monitoring, crowd management, and event evaluation.

1.4 Scope and Limitation

The study focused on the design and development of DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. The system was developed to improve the planning, management, monitoring, and evaluation of academic and extracurricular events by integrating RFID technology with artificial intelligence. It provided event organizers and administrators with intelligent analytics, predictive crowd monitoring, attendance validation, automated certificate generation, AI-assisted report generation, and event management tools. It also provided students and faculty with a centralized platform for event participation and engagement.

1.4.1 Scope

1.4.1.1 Users

1.4.1.1.1 Student and Faculty

Students and faculty members served as the primary users of the system. They were able to:

- Create an account and securely log in.
- Browse, search, save, and register for school events.
- View complete event information, including announcements, program flow, venue, schedule, organizers, and attendance requirements.
- Submit required documents during registration.

- View RFID attendance logs, attendance duration, and attendance validation status.
- Download and manage earned digital certificates.
- Participate in both mid-event and post-event surveys.
- Upload event photos when permitted by the event organizer.
- View their event participation history, certificates, attendance records, and account statistics.
- Receive notifications related to registrations, event updates, attendance, and certificates.
- Submit event-related and system-related feedback.
- Manage their account information, profile picture, and password.

1.4.1.1.2 Event Organizer

Authorized students or faculty members were able to organize events. In addition to the features available to regular users, organizers were able to:

- Create and submit event proposals.
- Save event drafts before submission.
- Monitor event approval status, including pending, approved, rejected, revision requested, postponed, cancelled, and draft events.
- Cancel or postpone approved events.
- Manage participant registration.
- Review and approve submitted participant requirements.
- Invite selected participants.
- Allow participants to upload event photos.
- Monitor event progress and participation.

1.4.1.1.3 Administrator

Administrators managed approved events and monitored event operations. Their responsibilities included:

- Reviewing, approving, rejecting, or requesting revisions for event proposals after venue validation through the iRoomReserve system.
- Monitoring RFID attendance in real time.
- Reviewing participant attendance records and performing manual attendance overrides when necessary.
- Monitoring AI-generated event analytics, such as predictive expected attendee count, attendance flow analysis, crowd density analysis, congestion prediction, crowd occupancy, registration trends, attendance prediction accuracy, and AI-generated attendance insights.
- Detecting duplicate RFID entry attempts, suspicious scanning behavior, invalid scans, and other attendance security alerts.
- Monitoring participant feedback collected during and after events.
- Generating, reviewing, and distributing digital certificates.
- Generating attendance, feedback, certificate, and AI-assisted event reports.
- Archiving event photos and generated reports for future reference.
- Managing student and faculty user accounts and monitoring their participation records.

1.4.1.1.4 Super Administrator

The Super Administrator had all administrator privileges, along with additional system management responsibilities, including:

- Managing administrator accounts.
- Creating, updating, suspending, and deleting administrator accounts.
- Assigning administrator permissions.
- Managing administrator activities and audit logs.
- Reviewing system-wide feedback.
- Monitoring administrator performance and account activity.
- Overseeing user management across all departments.
- Maintaining overall system administration and security.

1.4.1.2 System Modules

A. **Account and User Management Module** - Provided user registration, authentication, profile management, password management, role-based access control, and user account administration.

B. **Event Management Module** - Allowed organizers to create, update, postpone, cancel, and manage events while allowing administrators to review, approve, reject, or request revisions. The module was integrated with the eRoomReserve system for venue reservation validation before event approval.

C. **RFID Attendance Management Module** - Used RFID-enabled school ID cards for participant check-in and check-out, recorded attendance duration, validated attendance requirements, and maintained attendance history.

D. AI Event Intelligence Module - Used artificial intelligence to analyze event and attendance data and provided the following features:

- Predictive expected attendee count
- Attendance prediction
- Registration forecasting
- Attendance flow analysis
- Crowd density monitoring
- Congestion prediction
- Peak occupancy prediction
- AI attendance insights
- AI event performance analysis
- Recommendations for future events

E. Attendance Security Module - Detected duplicate RFID scans, suspicious scanning behavior, rapid consecutive scans, invalid scans, and manual attendance overrides to improve attendance reliability.

F. Certificate Management Module - Automatically validated attendance eligibility, generated digital certificates, distributed certificates, and monitored certificate generation and download status.

G. Feedback Management Module - Collected and managed both mid-event and post-event surveys, participant ratings, comments, and feedback analytics for completed events.

H. AI Report Generation Module - Automatically generated attendance reports, event reports, certificate reports, feedback reports, and comprehensive event reports with

AI-generated summaries, highlights, recommendations, and conclusions. It also allowed administrators to review and edit generated content before exporting.

I. **Notification Module** - Sent notifications regarding registrations, approvals, attendance validation, event updates, certificates, and other important system activities.

1.4.1.3 Software

A. **Next.js** – Served as the primary framework for developing the web application, including both frontend and backend services.

B. **MongoDB** – Served as the primary database for storing user information, event details, RFID attendance records, AI analytics, certificates, reports, feedback, and archived event data.

C. **Gemini AI** – Provided AI-powered attendance analysis, attendance prediction, crowd analytics, event insights, recommendations, and AI-assisted report generation.

D. **Figma** – Was used to design the user interface and user experience of the system before implementation.

E. **Visual Studio Code** – Served as the primary source code editor during system development.

F. **GitHub** – Was used for version control, collaboration, and source code management.

1.4.2 Limitation

The study has the following limitations:

1. The system was developed exclusively for St. Dominic College of Asia, including the Main Campus and Digital Campus.

2. The system only supported school-sponsored academic, organizational, and extracurricular events and was not intended for regular classroom attendance.
3. Venue reservation validation depended on the availability and successful integration of the eRoomReserve system.
4. Users were required to use their registered RFID-enabled institutional identification cards for attendance logging. Alternative attendance methods such as QR codes, biometrics, or facial recognition were outside the scope of the study.
5. AI predictions, including expected attendee count, attendance flow, crowd density, congestion prediction, and recommendations, were based on historical event data collected by the system and were considered decision-support information rather than guaranteed outcomes.
6. The accuracy of AI-generated reports and recommendations depended on the completeness and correctness of the collected attendance, event, and feedback data.
7. The system required a stable internet connection for real-time attendance synchronization, AI analytics, report generation, notifications, and cloud database access.
8. Only registered users with valid institutional accounts were permitted to access the system and participate in RFID attendance logging.
9. RFID authentication was limited to RFID cards compatible with the RFID reader used by the system, which operated at 13.56 MHz.
10. The study focused only on the development and evaluation of the proposed system and did not include deployment for institutions other than St. Dominic College of Asia.

1.5 Conceptual Framework

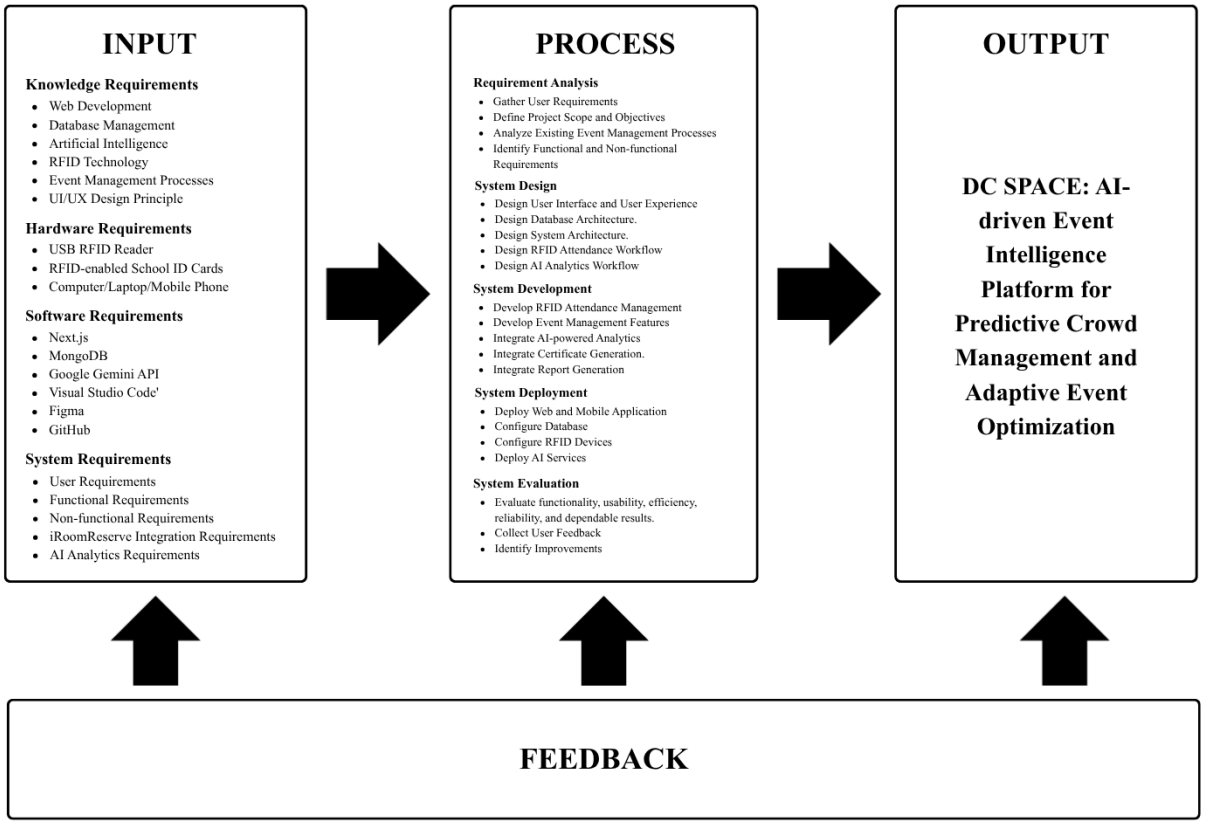


Figure 1.1 Input Process Output Framework of the Project

The conceptual framework of the study was based on the Input-Process-Output (IPO) model. The Input section consisted of the technical knowledge, software tools, hardware components, and system requirements that were utilized in developing the system. The Process section covered the different stages involved in the development of DC SPACE, which included requirement analysis, system design, development, testing, deployment, and evaluation. The Output section presented the completed DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. Lastly, the Feedback section involved gathering user and system feedback to identify possible improvements and determine whether the developed system met its intended purpose.

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1.6 Significance of the Study

This study, DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization, addresses a practical and recurring problem in educational institutions: the inefficiency of manual attendance and certificate distribution. Common issues such as delayed logging, illegible handwriting, duplicated entries, proxy attendance, and slow certificate release point to a clear need for a more automated and reliable approach to event administration, one that this study aims to provide. The findings of the study are expected to benefit the following:

- **Students**

  Students stand to benefit directly from a system that makes participating in school events faster and less of a hassle. Tapping an RFID card removes the need to manually sign in, cutting down on the delays that often build up during large seminars. Through their personal dashboard, students can also view their joined and organized events, track saved events, and monitor attendance and certificate status in one place all without having to follow up with organizers manually. Certificates, once earned, are available for download directly through the platform, making the process more convenient and accessible than waiting on manually distributed copies

- **Faculty**

  Faculty members also stand to benefit from the system, both as event participants and as advisers to student organizations. Like students, faculty can register for events, tap their RFID-enabled ID for attendance, and access earned certificates directly through their dashboard useful for documenting participation in training sessions, seminars, and other professional development activities. For faculty serving as

organization advisers, the system's centralized records also make it easier to stay informed about their organization's event submissions and attendance outcomes without needing to request updates manually from student officers. .

- **Administrators and Event Organizer**

For administrators and organizers, the study offers a more centralized way to manage the full lifecycle of an event from reviewing and approving submissions, to monitoring real-time attendance, to handling certificate issuance once requirements are met. Consolidating user management, event moderation, attendance tracking, and certificate handling into a single admin workflow reduces the back-and-forth that manual coordination usually requires. As the system's analytics capabilities mature, administrators are also expected to gain clearer visibility into participation trends and engagement patterns, supporting more informed decisions around event planning and resource allocation.

- **Educational Institutions**

At the institutional level, this study supports a broader shift toward digital, data-informed operations. Bringing together RFID technology, centralized data storage, and automated certificate generation gives the institution a more scalable way to manage the growing number of events, seminars, and activities it hosts each year. Moving away from paper-based sign-in sheets and printed certificates also reduces material costs and supports more sustainable event practices, while strengthening the accuracy and security of institutional records.

- **Researchers and Future Developers**

  Beyond its immediate use, this study can serve as a reference point for others working on RFID-based systems, event management platforms, or automated certification tools. The system's design decisions, feasibility analysis, and implementation approach including how attendance validation, notification handling, and certificate generation were addressed offer a practical starting point for future work in this space. As the system evolves toward more advanced features such as predictive analytics and deeper AI integration, this study may also help inform future research into smart campus technologies and similarly data-driven institutional systems.

1.7 Definition of Terms

1.7.1 Operational Terms

- **Event Attendance Monitoring:** The process of logging and tracking a participant's presence at a school event through RFID tap-in and tap-out actions, replacing manual attendance sheets with records that update automatically in real time.
- **Event Organizer:** The student organization, faculty member, department, or administrator authorized to create, manage, and oversee events within the DC Space system.
- **Certificate Distribution:** The automated process by which digital certificates are generated and released to participants who have satisfied the required attendance criteria for an event.

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- **Attendance Verification:** The process the system uses to confirm that a participant genuinely attended an event, based on RFID scan data and the resulting attendance logs.
- **RFID-Account Verification:** The safeguard that ties every RFID scan to the specific account it belongs to, ensuring that attendance can only be recorded by the actual ID holder and reducing the risk of proxy or fraudulent attendance.
- **Participant Registration Record:** The stored profile information of a participant including name, institutional ID, email, attendance history, and event participation maintained within the system's database.
- **Attendance Fraud Prevention:** The set of safeguards built into the system to reduce unauthorized attendance logging, proxy attendance, and duplicate entries by requiring RFID-based identity confirmation.
- **Automated Event Management:** The overall handling of institutional events through the DC Space system, covering event creation, attendance tracking, participant management, analytics, and certificate issuance.
- **Real-Time Attendance Tracking:** The system's ability to update and display attendance records immediately after an RFID scan occurs during an event.
- **Event Analytics:** The AI-supported insights the system generates from attendance and participation data, used to identify engagement trends, active organizations, and facility usage to guide institutional planning.
- **User Dashboard:** The web-based interface through which students, faculty, and administrators access event details, attendance records, certificates, and other system features relevant to their role.

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1.7.2 Technical Terms

- **RFID (Radio Frequency Identification):** A wireless identification technology that uses radio waves to detect and verify individuals through RFID-enabled ID cards and compatible readers.
- **RFID Reader:** The hardware device used to scan RFID-enabled cards and transmit the resulting attendance data to the DC Space system for processing.
- **Cloud-Based Database:** An internet-hosted, centralized database used to securely store attendance logs, user profiles, certificates, and event records.
- **Web-Based Application:** The DC Space platform itself, accessible through standard web browsers, allowing users and administrators to manage events, track attendance, and access certificates remotely.
- **Automated Email Notification System:** The system function responsible for sending automatic notifications and e-certificates to participants once their attendance has been validated.
- **System Development Life Cycle (SDLC):** The structured, phased methodology followed in building the DC Space system, encompassing planning, requirement analysis, design, development, testing, deployment, and maintenance.
- **Database Management System (DBMS):** The software layer responsible for organizing, storing, retrieving, and securing attendance records, participant data, and event information within the platform.
- **Artificial Intelligence (AI):** In the context of this system, the analytics layer is powered by Gemini AI that processes attendance and participation data to surface trends and support administrators in event-related decision-making.

- **Three-Tier Architecture:** A system design structured into three layers: presentation, application, and database that organizes how users, system logic, and stored data interact with one another.

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CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter reviews related literature and studies relevant to the development of DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. It covers RFID-based attendance, cloud computing and data management, digital certification, AI in event and crowd management, event technology, participant engagement, system architecture, and usability. These studies provide a foundation for understanding existing technologies and identifying gaps relevant to the development of an integrated event management system.

2.1 Artificial Intelligence in Event Management and Crowd Management

2.1.1 Artificial Intelligence in Event Management

The use of Artificial Intelligence (AI) became increasingly relevant in event management because of its ability to support planning, organization, information processing, and decision-making. Abdul Halim, Zamzuri, and Ghazali (2023) examined the transformative role of AI in the event management industry and discussed several applications and trends involving AI. Their study emphasized that AI had the potential to improve event-related processes while also presenting challenges that event organizations needed to consider when implementing the technology. The researchers highlighted the importance of using AI appropriately and considering its possible limitations and implications. Their findings demonstrated that AI could serve as a supporting technology in event management rather than completely replacing the role of event organizers. This

supported the use of AI in DC SPACE as a decision-support feature that helped organizers understand event information and make informed decisions.

Camacho-Ruiz et al. (2023) developed an artificial intelligence-based method for classifying tourism destination events. Their study used data science, supervised machine learning, and natural language processing techniques to automatically classify different types of tourism events. The researchers used BERT to obtain vector representations of event information and combined it with a classification model to organize events into a hierarchical taxonomy. They explained that events available online were often found in different and unstructured sources, which made manual classification time-consuming. Their proposed approach was able to organize event information into a more consistent catalog. Although the study focused on tourism destination events, it demonstrated how AI could process event-related information and organize it into useful categories. This was relevant to DC SPACE because the system managed different types of event information, including event details, registration information, attendance records, and event results.

Technology adoption in events also became an important part of improving event operations and experiences. Zamzuri et al. (2023) reviewed 28 studies related to event technology usage and experience. The reviewed studies covered event stakeholders' behavior, technological innovation, and experiences associated with using technology in events. The researchers emphasized that understanding how stakeholders used technology was important because the increasing use of technology could create both opportunities and challenges during events. Their findings showed that event technology should not only focus on introducing new technological features but should also consider how the technology was

experienced and used by different stakeholders. This was relevant to DC SPACE because the system was intended for participants, event organizers, administrators, and super administrators, who had different responsibilities and interactions with the platform.

Ryan et al. (2020) introduced the concept of "Events 4.0" to describe the digital maturity of events. Their study examined how different technologies could be placed along different levels of digital maturity. They explained that highly digitally mature events could integrate communication systems, optimize digital operations, and use technology to support event delivery, marketing, and customer experience. The study suggested that technology became more valuable when it was integrated into different parts of event operations instead of being used only for individual tasks. This concept was relevant to DC SPACE because the system integrated several event-related processes, including event management, registration, RFID attendance monitoring, feedback collection, certificate generation, reporting, and AI-based analysis within one platform.

The growing use of technology in events also supported the development of more integrated digital event systems. Causin and Scamacca (2021) discussed the role of technology in the meetings and events industry and explained that technological tools were used throughout different stages of the event planning lifecycle. Their discussion covered technologies used for pre-event planning, marketing and communication, on-site activities such as event badging and ticketing, virtual events, and post-event data collection and measurement. The authors emphasized that technology had become an important part of how meetings and events were planned, delivered, and evaluated. This was relevant to DC SPACE because the system integrated several event-related functions, including event management,

registration, RFID attendance monitoring, feedback collection, certificate generation, reporting, and AI-based analysis within one platform.

Based on these studies, AI and other digital technologies supported event management by organizing information, improving event operations, reducing repetitive work, and providing additional information for decision-making. The literature also showed that the effectiveness of event technology depended on how well it was integrated into event processes and how well it addressed the needs of its users. For DC SPACE, these findings supported the development of an integrated event management platform where AI served as a tool for providing insights and recommendations while event organizers and administrators remained responsible for making the final decisions.

2.1.2 AI-Based Crowd Management and Density Analysis

Crowd management was an important consideration when organizing events because a large number of participants within a limited space could result in congestion and other crowd-related concerns. Alasmari, Farooqi, and Alotaibi (2024) conducted a systematic literature review of deep-learning techniques used in crowd management. Their review examined 45 primary studies and identified different applications of deep learning in crowd management, including crowd counting, scene analysis, crowd behavior detection, and prediction. The researchers explained that deep-learning techniques could help analyze crowd conditions and support strategies for managing large gatherings. However, they also identified limitations related to datasets, hardware requirements, computational costs, and the ability of models to generalize to new crowd patterns. These findings demonstrated both the potential and limitations of AI-based crowd management.

Xiang (2022) developed a crowd density estimation method using deep learning for a passenger-flow detection system in an exhibition center. The study focused on problems involving crowd distribution, differences in crowd scale, and the difficulty of extracting useful crowd features from exhibition environments. The proposed method used a convolutional neural network-based approach to estimate crowd density and detect passenger flow. The study demonstrated that deep learning could be applied to analyze crowd density in exhibition centers, where a large number of people gathered within a limited area. This was relevant to DC SPACE because school events could also involve many participants gathering in specific venues, making information about participant distribution useful for event organizers.

Bhuiyan et al. (2021) examined crowd density estimation using deep learning for Hajj pilgrimage video analytics. The researchers focused on high-density crowds and the use of video analysis to support crowd control and safety in large gatherings. Their work demonstrated how automated crowd-density estimation could provide information about crowded areas that might be difficult to monitor manually. Although the study focused on Hajj pilgrimage crowds, the concept could also be applied to other environments where large numbers of people gathered. For DC SPACE, this supported the use of automated data analysis to provide information about crowd conditions during events.

Khan, Menouar, and Hamila (2023) developed LCDnet, a lightweight crowd-density estimation model for real-time video surveillance. Their study addressed the increasing need for automatic crowd counting and density estimation and developed a lightweight model

intended for real-time applications. The researchers explained that crowd counting and density estimation had become important applications of computer vision and deep learning. Their work demonstrated that AI models could be designed to automatically estimate crowd conditions while considering the requirements of real-time monitoring. This supported the concept of using automated analysis to assist event personnel in monitoring crowd conditions when manually observing the entire venue was difficult.

Yamin et al. (2023) proposed a deep-learning approach for classifying different levels of crowd density. Their study developed the Sparrow Search Optimization with Transfer Learning-Based Crowd Density Classification model, which used an EfficientNet model to extract features and classify crowd-density conditions. The researchers evaluated the approach using datasets representing different crowd densities and reported a maximum classification accuracy of 93.25%. They explained that crowd-density classification could be useful for managing Hajj and other crowded events. This demonstrated how AI could move beyond simply counting people and could also classify the condition of a crowd. For DC SPACE, this supported the use of AI-generated crowd-density information to help organizers understand whether an event area might be experiencing low, moderate, or high crowd conditions.

The reviewed studies showed that AI-based crowd analysis could be used to estimate crowd density, count people, analyze participant distribution, and classify different crowd conditions. However, many existing approaches relied on video datasets, cameras, or other sources of data that required sufficient equipment and suitable training data. Alasmari et al. (2024) specifically identified limitations involving dataset availability, computational

requirements, hardware, and model generalizability. Therefore, the use of AI for crowd management needed to consider the availability and quality of the data used by the system.

For DC SPACE, these findings supported treating AI-generated crowd and attendance insights as decision-support information rather than guaranteed predictions. The system used available registration and RFID attendance data as sources of information for estimating expected attendance, attendance flow, crowd density, and possible congestion. Since the accuracy of these predictions could depend on the amount and quality of historical event data collected by the system, organizers and administrators needed to interpret the generated results together with the actual conditions of the event.

2.1.3 AI and Data-Driven Event Decision Support

AI became more useful in event management when the information it produced could be used to support decisions. Abdul Halim et al. (2023) discussed the use of AI in event management, while Camacho-Ruiz et al. (2023) demonstrated how AI could process and classify event information. Similarly, Alasmari et al. (2024), Xiang (2022), Khan et al. (2023), and Yamin et al. (2023) demonstrated how deep-learning methods could analyze different crowd conditions. Together, these studies showed that AI could process event-related information and produce outputs that could assist organizations in understanding situations that might be difficult to evaluate manually.

In DC SPACE, this concept was applied by using information collected from different parts of the system as inputs for AI-based analysis. Event registration information provided an indication of expected participation, while RFID attendance records provided information

about actual participant entry and exit. These data contributed to analyses involving expected attendee count, attendance flow, crowd density, congestion prediction, and recommendations. The purpose of these features was to provide organizers with additional information that could help them identify possible concerns before or during an event.

The use of AI as decision support also addressed an important limitation of predictive systems. AI-generated results could vary depending on the quality, quantity, completeness, and relevance of the available data. The literature on crowd-management AI also identified concerns involving dataset limitations and model generalizability. Therefore, the predictions generated by DC SPACE were not treated as exact or guaranteed outcomes. Instead, they were intended to provide additional information that organizers and administrators could consider together with actual event conditions, observations, and their own judgment.

2.2 Cloud Computing in Information Systems

2.2.1 Cloud-Based Information Systems

Cloud computing has increasingly become a preferred infrastructure choice for institutional information systems because it allows organizations to deliver IT services over the internet rather than maintaining costly on-premise servers. Research on cloud adoption in higher education institutions has framed this shift through an integrated theoretical lens, showing that adoption decisions are driven by perceived usefulness, ease of use, and the institution's readiness to migrate existing services to cloud infrastructure (Al-Sharafi et al., 2021). A related study on universities transitioning to cloud computing found that budget limitations, licensing costs, and the burden of hardware/software maintenance are common

barriers that cloud migration directly addresses, since it shifts infrastructure responsibility away from the institution and toward the cloud provider (Aydin, 2021). Together, these findings suggest that cloud-based information systems are not simply a technical upgrade but a response to real operational and financial constraints faced by schools and universities.

This pattern applies directly to DC Space, which is deployed as a cloud-hosted Next.js application rather than a system requiring dedicated institutional servers, a design choice consistent with the resource constraints these studies describe for academic institutions.

2.2.2 Cloud Storage and Data Accessibility

A recurring theme in cloud storage literature is the tension between data accessibility and data security systems need to make records available quickly while still protecting them from unauthorized access. Dayana and Rani (2023) proposed a cloud storage approach combining Keccak hashing and AES encryption specifically to reduce the time and memory overhead of securing stored data, demonstrating that accessibility and protection do not have to be mutually exclusive design goals. Their findings reinforce that as institutions move records to the cloud, the underlying storage method significantly affects how quickly authorized users can retrieve information without compromising its integrity.

In DC Space, this trade-off shows up in how attendance logs, certificates, and user records are stored in MongoDB rather than local files centralizing data so it can be retrieved on demand while still requiring authenticated access through JWT and NextAuth sessions.

2.2.3 Cloud Integration in Automated Systems

Cloud infrastructure is what makes real-time automation possible in attendance and event-tracking systems, since it removes the delay between data capture and data availability. Naen et al. (2021) developed a biometric attendance system that pairs a fingerprint scanner with cloud-based storage and AI-assisted processing, and found that routing attendance data through the cloud immediately after capture allowed the system to generate reports (daily, weekly, monthly attendance) without the batch-processing delay that manual or offline systems require. Their findings support a broader pattern in cloud-integrated attendance research: the capture technology can vary fingerprint, RFID, facial recognition but it's the cloud connection specifically that makes real-time reporting and analytics feasible.

The same principle applies in DC Space where the choice of RFID as the capture method is secondary to the fact that tap-in/tap-out events are transmitted to a cloud-hosted application layer, which is also where the Gemini AI analytics endpoint processes attendance data.

2.2.4 Benefits of Cloud Computing in Educational Institutions

Bhardwaj et al. (2021), examining cloud computing adoption across Indian public universities during the COVID-19 period, found that cloud services allowed institutions to rapidly extend access to students and staff without proportional increases in physical infrastructure, and that cost-efficiency was a major factor in sustained adoption after the initial rollout. Separately, Alshamaila et al. (2020) found in the context of SME adoption that cost, relative advantage, and top management support were the strongest predictors of successful cloud adoption, a pattern that mirrors what smaller institutions face when weighing cloud investment.

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These findings align with DC Space's own economic feasibility analysis, where cloud hosting was chosen precisely because it avoided the cost of physical servers while still allowing the system to scale across school-wide events.

2.3 Event Technology, Feedback, Evaluation, and Participant Engagement

2.3.1 Event Technology and Participant Engagement

Technology was increasingly integrated into events to support communication, participation, event operations, and attendee experience. Chen and Jung (2024) conducted a systematic review of event technology and attendee experience using the PRISMA approach. Their study noted that although event-related technologies had been widely researched, there was still a need to understand the specific roles that technology played in facilitating attendees' experiences. The review emphasized that event technology should not only be evaluated according to its operational functions but also according to how attendees interacted with and experienced the technology. This was relevant to DC SPACE because participants were among the primary users of the system and interacted with it through different stages of an event.

Zamzuri et al. (2023) similarly reviewed literature concerning event technology usage and experience and identified themes involving stakeholder behavior, technology innovation, and user experience. The researchers reviewed 28 articles and emphasized the importance of understanding how different stakeholders used technology during events. Their findings indicated that event technology needed to be designed with consideration for both operational needs and the experiences of the people who used it. This was relevant to DC SPACE

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because participants, event organizers, administrators, and super administrators had different roles and therefore interacted with different functions of the system.

Ramli et al. (2023) examined factors influencing technology usage among event attendees. The researchers collected 202 responses from event-management students and investigated factors such as memorable experience, perceived value, enjoyment, and technology usage. Their study showed that participants' experiences and perceived value could be associated with their use of event technology. This suggested that event systems needed to provide functions that participants found useful and meaningful instead of simply transferring manual event processes into a digital format. For DC SPACE, this supported the importance of providing participants with useful functions such as event browsing, registration, attendance recording, surveys, notifications, and access to certificates.

Neuhofer, Magnus, and Celuch (2021) examined the potential impact of artificial intelligence on event experiences using a scenario-based approach. Their study explored how AI could influence different aspects of event experiences and discussed the opportunities and challenges that could arise from the increasing use of AI in events. The study highlighted that AI could influence not only the operational side of events but also how attendees experienced and interacted with event environments. This was relevant to DC SPACE because the system used AI as part of its event intelligence features while also providing participants with digital functions such as event information, registration, attendance monitoring, feedback, and certificates. The study therefore supported the consideration of both technological functionality and attendee experience when integrating AI into event systems.

The reviewed studies showed that event technology could support participants throughout different stages of event participation. The usefulness, ease of use, perceived value, and overall experience provided by the technology could influence how participants interacted with it. For DC SPACE, these findings supported the development of a centralized platform where participants could access event information, register for activities, record attendance through RFID, provide feedback, receive notifications, and access digital certificates. Combining these functions within one system helped reduce the need for participants to use separate processes for different stages of their event participation.

2.3.2 Digital Feedback and Event Evaluation

Feedback was an important part of event evaluation because it provided organizers with information about participants' experiences. Chen and Jung (2024) emphasized the importance of considering attendees' experiences when studying event technology. Their systematic review showed that event technologies could influence how participants experienced and interacted with events. This provided a basis for considering digital feedback and evaluation as part of an integrated event technology system rather than treating feedback as a separate activity.

Selman et al. (2023) evaluated the Good Grief Festival using participant feedback collected through online surveys. The evaluation included surveys conducted before and after the festival to examine participants' experiences and the impact of the event. The study demonstrated how digital surveys could be used to gather information from participants and evaluate an event's reach and impact. This was relevant to DC SPACE because the system

provided digital feedback mechanisms that could collect participant responses and store them as part of an event's records.

Brescia-Zapata (2023) examined approaches for organizing sustainable and accessible events through co-creation workshops and applied the findings to the organization of the Green Digital Accessibility conference. The study included audience feedback as one of the steps in evaluating the effectiveness of the measures implemented during the event. The researchers also collected participant feedback through a post-event questionnaire after the conference. This demonstrated that post-event questionnaires could be used to gather participant responses and identify areas that could be considered when organizing future events. For DC SPACE, this supported the use of digital post-event surveys as part of the event evaluation process.

For DC SPACE, these findings supported the inclusion of both mid-event and post-event surveys. Mid-event surveys allowed organizers to collect participant responses while an event was still taking place, while post-event surveys were used to evaluate the overall experience after the event. Ratings, comments, and survey responses could be stored in the system and reviewed by authorized event organizers and administrators. This allowed participant feedback to become part of the event's collected information and provided organizers with another source of information when evaluating event performance.

2.3.3 Technology-Supported Event Engagement Across the Event Lifecycle

Event engagement did not only occur during the actual event. It could begin before the event through registration and access to event information and continue after the event

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through feedback and future participation. Chen, Jung, and Cai (2025) conducted a critical review of technology-facilitated event engagement and analyzed 79 articles concerning technology use in different event contexts. Their study examined how technology supported attendee engagement during the pre-event, during-event, and post-event stages. The researchers emphasized that technology-facilitated engagement could occur throughout the event lifecycle instead of being limited to the actual event period.

This lifecycle perspective was closely related to DC SPACE. Before an event, participants could browse available events, view event information, and register for activities. During the event, RFID attendance records were used to record participant entry and exit, while surveys were used to collect participant responses. After the event, participants could provide feedback, review their attendance information, and receive digital certificates when they met the required attendance conditions. In this way, the system supported participant interaction across different stages of an event.

The integration of these activities also allowed DC SPACE to collect information from different stages of the event. Registration data provided information about expected participation, RFID attendance records provided actual attendance information, and feedback provided information about participant experiences. These data could then be incorporated into event reports and used as inputs for AI-assisted analysis. This made participant engagement and feedback not only features for users but also sources of information that could contribute to event evaluation and future planning.

2.3.4 Feedback as a Basis for Event Improvement

Participant feedback became more useful when the information collected could be reviewed and used to improve future events. Chen, Jung, and Cai (2025) emphasized that technology-facilitated event engagement could provide opportunities to understand attendee interactions and experiences across the event lifecycle. Their review highlighted the potential of technology to support event engagement and improve how organizers understood and responded to attendee experiences.

Ramli et al. (2023) also found that participant experiences and perceived value were related to the use of event technology. This suggested that understanding how participants experienced an event and the technologies used during the event could help organizers determine whether the technology provided value to attendees. Therefore, collecting participant feedback provided additional information that could be considered when evaluating event performance.

For DC SPACE, participant feedback served as one of the sources of information for event improvement. Ratings and comments collected through the system could be reviewed by organizers and administrators and could also be incorporated into generated event reports. When combined with attendance records and other event statistics, feedback provided a more complete view of event performance. The collected information could then help organizers identify areas that needed improvement and serve as a reference when planning future events.

2.3.5 Integrated Event Technology and Event Engagement

The literature reviewed in this cluster showed that event technology supported both operational processes and participant experiences. Chen and Jung (2024) emphasized the importance of attendee experience when examining event technology, while Zamzuri et al. (2023) highlighted stakeholder behavior, technology usage, and event experience. Ramli et al. (2023) further demonstrated that perceived value, usefulness, ease of use, and other factors could influence participants' use of event technologies. These findings suggested that an effective event system needed to not only automate event processes but also provide useful and accessible functions for its users.

2.4 Database Management & System Architecture

2.4.1 Database Management Systems (DBMS)

Database management systems remain a foundational concern in software engineering research, particularly around how well different DBMS technologies perform under real-world conditions rather than idealized benchmarks. Taipalus (2024) conducted a systematic literature review of DBMS performance comparison studies and found that most published performance benchmarks fail to reflect real-world usage patterns, and that many studies report results in insufficient detail to be replicated or generalized. This finding is significant for system designers: it suggests that choosing a DBMS should be grounded in the specific operational profile of the system being built (read-heavy vs. write-heavy, structured vs. flexible data, concurrent access patterns) rather than generic performance claims.

This is directly relevant to DC Space's choice of MongoDB as its primary datastore. Since the system's core operations RFID tap-in/tap-out logging, real-time dashboard updates,

certificate generation involve frequent writes and flexible, evolving data structures (event fields, attendance metadata), the DBMS choice should be evaluated against that specific access pattern rather than assumed.

2.4.2 Database Design for Attendance Systems

Research directly comparing relational and non-relational (NoSQL) databases offers useful guidance for attendance-style systems specifically. Jose and Abraham (2020) compared MongoDB and MySQL performance across varying dataset sizes and query types, finding that non-relational engines with MongoDB performing best handled large-volume operations more efficiently than relational databases, though MongoDB required more memory and CPU resources to do so. This trade-off matters for attendance systems, which generate a continuously growing volume of timestamped records (tap-in/tap-out events) that must be written and retrieved quickly during live events.

DC Space's database design reflects this pattern directly; attendance data is split across attendance_logs and attendance collections in MongoDB, supporting the kind of high-frequency, flexible-schema writes that Jose and Abraham found MongoDB handles well, at the cost of the higher resource usage their study also identified.

2.4.3 System Architecture for Information Systems

Three-tier architecture remains the standard approach for structuring web-based institutional systems, separating presentation, application logic, and data storage into independently manageable layers. Barrocan et al. (2025) developed a three-tier web-based monitoring system for tracking internship attendance at a Philippine state university, and found that the architecture's separation of concerns directly supported the system's scalability, flexibility, and secure data management allowing individual layers to be updated or

maintained without disrupting the others. Their findings also showed measurable reductions in human error and administrative workload once the manual, paper-based tracking process was replaced by the three-tier system.

This maps closely onto DC Space's own architecture: the presentation tier (Next.js frontend and RFID hardware interface), the application tier (API routes, Gemini AI analytics, certificate generation logic), and the data tier (MongoDB collections) mirror the same separation Barrocan et al. describe and for much the same institutional attendance-tracking purpose.

2.4.4 Data Security and Data Integrity

Data integrity is a distinct concern from data security in RFID-based systems; specifically a system can restrict who accesses data while still being vulnerable to the data itself being tampered with or cloned. Popa et al. (2026) addressed this directly in a university campus deployment, proposing a checksum-based data-integrity verification scheme for RFID cards that detects unauthorized modification of stored card data, tested over 10,000 authentication attempts with a 100% rejection rate for tampered or cloned cards. Their work highlights that RFID systems need integrity verification as a distinct layer from authentication confirming a card is genuine is not the same as confirming its stored data hasn't been altered.

This distinction is relevant to DC Space's current security model, which relies on RFID-to-account matching, OTP verification, and institutional email authentication for access control but data-integrity verification of the kind Popa et al. describe is a layer worth

considering as a documented limitation or future enhancement, since it addresses a different threat than authentication alone.

2.5 Human-Computer Interaction (HCI) & Usability

2.5.1 UI/UX Design Principles

User-centered design has long been established as the standard framework for building interfaces that genuinely serve their users rather than just their designers. Gulliksen et al. (2003) outlined twelve key principles for user-centered systems design, including the need for active user involvement throughout development, iterative design informed by real user feedback, and interfaces that match users' actual work practices rather than an idealized version of them. Their work remains widely cited because it treats usability not as a final polish step but as a requirement threaded through the entire design process from early requirements gathering through post-deployment evaluation.

This framing is relevant to DC Space's dashboards, where features like search filters, summary cards, and categorized event sections were shaped around actual user workflows (browsing events, tracking attendance, managing certificates) rather than added as an afterthought.

2.5.2 Kiosk Interface Design

Public-facing, walk-up interfaces carry different usability demands than personal devices, since users typically interact with them briefly, without training, and often under time pressure. Lee et al. (2025) evaluated the usability of self-ordering kiosk interfaces, testing multiple design variants using the System Usability Scale and Net Promoter Score,

and found that interfaces prioritizing focused information display and reduced customization complexity produced significantly fewer user errors than more feature-dense alternatives. Their findings reinforce a recurring theme in kiosk-specific HCI research: simplicity and error-reduction matter more at a walk-up terminal than the feature completeness that might be appropriate for a personal dashboard.

This is directly applicable to DC Space's RFID attendance points, where the interaction (tap card, receive confirmation) is intentionally minimally consistent with Lee et al.'s finding that reduced complexity lowers error rates in public, brief-interaction contexts.

2.5.3 Technology Acceptance Model (TAM)

Understanding whether users will actually adopt a new system regardless of its technical capability is a persistent concern in information systems research, and TAM-derived models remain the dominant framework for studying it. Xue et al. (2024), in a systematic review of UTAUT applications across higher education, found that performance expectancy consistently exerts the strongest influence on behavioral intention to adopt a new system, and that the effects of UTAUT factors vary meaningfully between higher education and general institutional contexts. In a more directly comparable context, Kamis et al. (2025) applied TAM specifically to electronic certificate systems and found that perceived usefulness, particularly the time saved compared to manual certificate processes, was the strongest predictor of user acceptance, more so than ease of use alone.

This has a direct implication for DC Space: since e-certificate issuance is one of the system's core value propositions, Kamis et al.'s finding suggests that user acceptance may

hinge more on participants perceiving a genuine time/effort savings over manual certificate distribution than on interface polish alone.

2.5.4 Usability Evaluation Methods

Rigorous usability evaluation requires selecting methods appropriate to what is actually being measured; a single technique rarely captures the full picture. Maqbool and Herold (2023), in a systematic review of 610 usability evaluation studies, found that inquiry-based methods such as questionnaires and interviews were used far more frequently than testing or inspection methods, underscoring how often method selection is driven by convenience rather than by what the evaluation actually requires. Usability testing allows developers to see how students, instructors, and administrators engage with system interfaces, RFID attendance modules, and credential administration capabilities. This enables the researchers to detect usability concerns such as navigational challenges, system delays, or user confusion throughout attendance procedures.

This bears directly on a limitation worth acknowledging in DC Space: the current feedback mechanism collects ratings and comments through the Submit Feedback form, but this data does not yet persist to the API meaning the kind of deliberate, method-appropriate evaluation Maqbool and Herold describe isn't fully possible for that channel until the persistence gap is closed.

CHAPTER 3

METHODOLOGY

3.1. Project Design Outline

3.1.1 System Objectives

The developed system, DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization, served as a web-based event management platform that integrated RFID technology and artificial intelligence to improve event planning, attendance monitoring, crowd management, participant engagement, and post-event evaluation at St. Dominic College of Asia.

Specifically, the system achieved the following objectives:

A. User Management and Access Control

- Developed a secure authentication system that allowed students, faculty members, event organizers, administrators, and super administrators to access the platform according to their assigned roles.
- Provided user profile management features, including account registration, profile updates, password management, and account statistics.
- Implemented role-based access control to ensure that users could only access the features and information appropriate to their responsibilities.

B. Event Management

- Developed an event management module that allowed organizers to create, edit, submit, postpone, cancel, and manage school events.
- Implemented participant registration, document submission, participant invitation, and registration approval features.
- Integrated the system with the iRoomReserve platform to validate venue reservations, schedules, and venue capacities before event approval.

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- Provided comprehensive event information, including announcements, program flow, attendance requirements, venue details, and organizing departments.

C. RFID Attendance Management

- Developed an RFID-based attendance monitoring system that recorded participant check-in and check-out using institutional RFID identification cards.
- Monitored attendance duration and validated whether participants met the required attendance time for certificate eligibility.
- Provided real-time attendance records and attendance summaries for participants, organizers, and administrators.

D. AI-driven Event Intelligence

- Integrated artificial intelligence to predict expected attendee counts before an event based on available event and registration data.
- Analyzed attendance flow to identify participant arrival and departure trends throughout an event.
- Monitored venue occupancy by analyzing crowd density and predicting possible congestion during the event.
- Generated AI-powered insights and recommendations that assisted organizers and administrators in improving event planning, resource allocation, and crowd management.
- Provided predictive analytics that supported adaptive decision-making before, during, and after school events.

E. Attendance Security Monitoring

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- Developed attendance security features that detected duplicate RFID entry attempts, suspicious scanning behavior, invalid scans, rapid consecutive scans, and manual attendance overrides.
- Notified administrators of potential attendance irregularities to improve the integrity and reliability of attendance records.

F. Certificate Management

- Developed an automated certificate generation module that validated participant attendance and generated digital certificates for qualified attendees.
- Allowed participants to securely view, download, and manage their digital certificates.
- Notified eligible participants when certificates became available.

G. Feedback and Participant Engagement

- Provided mid-event and post-event surveys to collect participant evaluations and suggestions.
- Allowed participants to submit event-related and system-related feedback.
- Enabled organizers to review survey results and participant feedback to support continuous event improvement.

H. AI-assisted Report Generation

- Developed an automated report generation module that compiled attendance records, participant feedback, event statistics, certificate summaries, and archived event media into comprehensive event reports.
- Utilized artificial intelligence to generate event summaries, key highlights, recommendations, and evaluation insights that could be reviewed and edited by administrators before finalization.

- Archived completed reports and event documentation for future reference and institutional recordkeeping.

I. Administrative and System Management

- Developed administrative modules that allowed administrators to manage users, monitor attendance, review event proposals, approve certificates, monitor analytics, and generate reports.
- Developed a Super Administrator module that managed administrator accounts, monitored administrator activities, reviewed system feedback, maintained audit logs, and oversaw overall system administration.
- Implemented notification features that informed users of event updates, attendance validation, registration status, certificate availability, and other important system activities.

3.1.2 System Framework

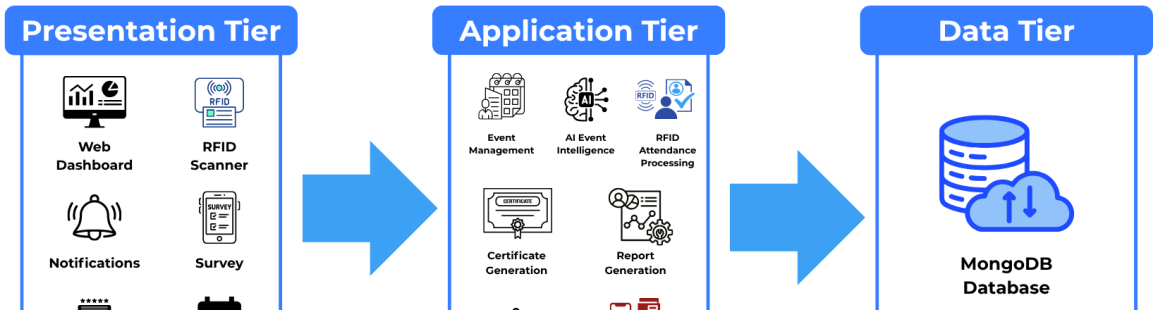


Figure 3.1. Three-Tier Architecture of the Proposed System

DC SPACE utilized a Three-Tier Architecture to support efficient school event management. The Presentation Tier served as the interface where users interacted with the system through a web dashboard. Through this layer, users were able to register for events, receive notifications, submit surveys and feedback, and monitor event activities. Participants also used RFID scanners to record their attendance during events. The Application Tier served as the core processing layer of the system. It handled event management, RFID attendance processing, AI event intelligence, certificate generation, report generation, notification services, and integration with the eRoomReserve system for venue validation. Lastly, the Data Tier served as the centralized storage layer of the system. It utilized MongoDB to store user information, event records, attendance logs, certificates, reports, feedback, and other system-related data. This layer supported reliable data management and the overall operations of DC SPACE.

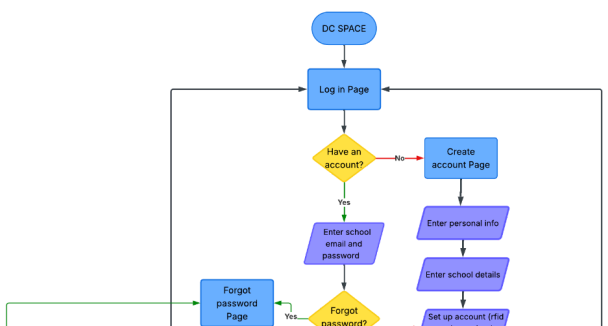


Figure 3.2 User Dashboard Flowchart

The User Flowchart shows how students and faculty members use the DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. The process starts when the user logs in using their school email and password. If the user does not have an account, they can create one by entering their personal and school information, verifying their email, and agreeing to the consent form. If the user forgets their password, they can use the Forgot Password feature to create a new one.

After logging in, users can browse and join events, upload required files, view event details, check their RFID attendance logs, download certificates, receive notifications, answer

surveys, submit feedback, manage their profile, and change their password. Users can also save events they are interested in and view their attendance and certificate history.

If the user has Event Organizer access, they can also create and manage events. They can create event proposals, save drafts, view pending and inactive events, review participant submissions, invite participants, monitor attendance, and manage their organized events. When they are done using the system, they can log out securely.

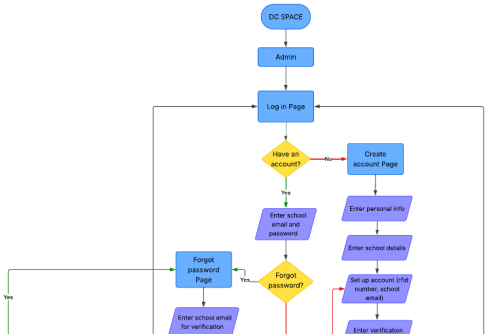


Figure 3.3 Administrator System Flowchart

The Admin Flowchart shows how administrators use the DC SPACE system to manage school events and monitor system activities. The process starts when the administrator logs in using their school email and password. They can also create an account or reset their password when needed. After logging in, administrators can access the Dashboard, Events, Attendance, Certificates, Feedback, Reports, and Notifications pages.

They can review events, monitor attendance, manage certificates, view event-related feedback, and generate reports.

Administrators can also manage student and faculty accounts, but they cannot make a user an administrator. They also cannot view system-related feedback. The Administration Page is not available to them. They can update their profile and account settings before logging out.

The administrator does not have access to the Administration Page, as this page is exclusively for managing administrator accounts. In the My Profile page, the administrator can view and update their personal information and account settings, including information about the Super Administrator who approved their administrator account. After completing their tasks, the administrator can securely log out of the system.

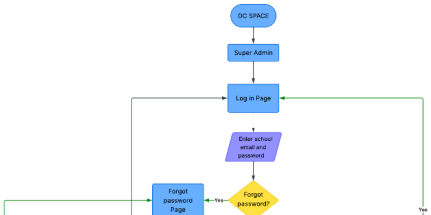


Figure 3.4 Super Administrator System Flowchart

The Super Admin Flowchart shows how the Super Administrator manages the DC SPACE system and administrator accounts. The process starts when the Super Administrator logs in using their school email and password. Unlike administrators, they do not have a Create Account option but can reset their password when needed. The Super Administrator can access the Dashboard, Events, Attendance, Certificates, Feedback, Reports, and

Notifications pages, similar to the Administrator. They can also manage student and faculty accounts and make a user an administrator.

The Super Administrator can view both event-related and system-related feedback and update the status of system issues. They also have exclusive access to the Administration Page, where they can manage administrator accounts and set their System Permissions. They can also manage their profile and securely log out of the system.

3.1.3 Feasibility Studies

3.1.3.1 Technical Feasibility

Technical feasibility evaluates whether the required technologies, hardware components, and software tools needed to develop and implement the proposed system are available and capable of supporting the project. The proposed system, DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization, is a web-based platform designed to automate attendance tracking during seminars and events.

The system uses RFID cards to identify participants, and the scanned data from the RFID device is transmitted to the web-based system, where attendance records are stored and processed. Administrators and event organizers can monitor attendance in real time and automatically generate certificates for qualified participants. Since the required technologies such as RFID readers, web development frameworks, and cloud hosting platforms, are widely available, the implementation of the proposed system is considered technically feasible.

3.1.3.1.1 Hardware Requirements

Minimum Hardware Requirements for System Development

The development of the web-based system requires a computer capable of running development tools, programming environments, and local servers necessary for building the application. The development team will use computer systems with at least the following specifications:

- **Processor**: Intel Core i5 or equivalent
- **Memory (RAM)**: Minimum of 8 GB RAM
- **Storage**: At least 256 GB SSD or HDD
- **Operating System**: Windows 10 or higher, macOS, or Linux
- **Internet Connection**: Stable internet connection for cloud deployment and development tools

These specifications are sufficient for running software development tools such as code editors, package managers, IoT development environments, and local testing servers.

Minimum Hardware Requirements for End-Users

The system will be accessible through a web browser, allowing administrators, event organizers, and participants to interact with the system using different devices. Since the system is web-based, users do not need to install specialized software.

Minimum requirements for end-users include:

- Desktop computer, laptop, tablet, or smartphone
- At least 4 GB RAM recommended
- Updated web browser such as Google Chrome, Microsoft Edge, or Safari
- Stable internet connection

These requirements ensure that users can access dashboards, monitor attendance data, and manage events through the web interface without performance issues.

3.1.3.1.2 IoT Hardware Components

The system will utilize several electronic components that enable automated attendance logging and event access control.

- **USB RFID Reader –** The USB RFID reader will be used to scan RFID cards assigned to participants during events. The device connects directly to a computer through a USB connection and reads RFID cards operating at a frequency of 13.56 MHz. It allows the system to capture tap-in and tap-out attendance data accurately and in real time.
- **RFID Cards (HF 13.56 MHz)** - High-frequency RFID cards will serve as identification tags assigned to participants. Each card contains a unique identifier that will be linked to a participant's student ID and profile in the system database.
- **USB Cable** - A USB cable will be used to program the ESP32 microcontroller and supply power during development and testing.

3.1.3.1.3 Software Requirements

Programming Languages

- **JavaScript** - JavaScript will be the primary programming language used in developing the web application. It will manage application logic, user interactions, and communication between the client interface and the backend services.
- **HTML and CSS** - HTML and CSS will be used to structure and design the user interface of the web system. These technologies will allow the development of

interactive dashboards for administrators, organizers, and participants while ensuring responsive design across different devices.

3.1.3.1.4 Frameworks

- **Next.js** - Next.js will be used as the primary full-stack web framework for building the system. It enables the development of both frontend and backend functionalities within a single project. The framework will manage user authentication, API routes, server-side processing, and dashboard interfaces.
- **Tailwind CSS** - Tailwind CSS will be used to design responsive and visually consistent user interfaces. It helps streamline the development process by providing utility-based styling for building dashboards and system pages.

3.1.3.1.5 Platforms and Services

- **Vercel (Web Hosting Platform)** - The system will be deployed using a cloud hosting platform such as Vercel. This platform is optimized for Next.js applications and provides reliable infrastructure for hosting the web system. Deploying the system online enables administrators and organizers to access the platform and monitor attendance records in real time.
- **MongoDB** - MongoDB will serve as the database system responsible for storing event information, participant profiles, RFID attendance records, and certificate-related data. It allows the system to manage and retrieve data efficiently while supporting real-time updates and flexible data storage for the web-based application.
- **RFID Reader** - The RFID reader will serve as the hardware device used to scan RFID cards or IDs of participants during events. It captures attendance data through

tap-in and tap-out transactions and sends the information to the system for real-time attendance monitoring and recording.

3.1.3.1.6 Data Requirements

The system will store various types of data necessary for managing events and monitoring attendance.

- **Login Data** - Authentication information such as email addresses, encrypted passwords, and user roles will be stored to control system access.
- **User Profile Data** - Profiles will contain personal details including name, email address, and participant or organizer roles.
- **Student ID Information** - Student ID numbers will be linked to RFID card identifiers to uniquely identify participants when they scan their cards.
- **Attendance Records** - Attendance logs will record the event ID, participant ID, date, and time of each RFID scan. This allows the system to track participant attendance accurately.
- **Analytics Data** - The system may generate analytics such as attendance summaries, participation rates, and event statistics to assist administrators in evaluating event participation.

3.1.3.2 Operational Feasibility

The operational feasibility of the proposed RFID-based attendance system was evaluated based on responses from DominiXode, Dominican Nursing Student Council, Junior Travelers Tourism Council, and Laboratorium Duces.

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The results show that the current manual attendance system is inefficient, as it is time-consuming, prone to delays during large events, and affected by issues such as unreadable handwriting, incomplete information, duplicate entries, proxy signing, and lack of real-time tracking and proper verification of attendance duration. In contrast, the majority of organizations strongly agree that the RFID-based system would improve attendance tracking and that automating certificate generation would be beneficial. Most respondents also agree or strongly agree that the system will make their tasks easier, reduce workload, improve efficiency, and enhance event coordination through real-time monitoring.

While users are generally willing to learn and adopt the system, some concerns remain, including unfamiliarity with the technology, possible technical issues, data privacy, and usability challenges. To address this, respondents emphasized the need for basic orientation, hands-on training, user guides, and troubleshooting support.

Despite expected challenges during the transition, such as user adaptation and system setup, the organizations believe that the new system will make workflows more efficient, reduce administrative tasks, and improve overall event management. Therefore, the proposed system is considered operationally feasible, as most organizations are willing to adopt it and recognize its potential benefits.

3.1.3.3 Economic Feasibility

The economic feasibility of DC SPACE was evaluated by considering the estimated development costs and expected benefits of the system. The assessment determined whether the projected savings from automating event management, attendance monitoring, certificate

generation, reporting, and other related processes were sufficient to justify the estimated initial investment.

The project's estimated initial investment amounted to ₱4,000, which mainly covered the hardware components needed for RFID attendance monitoring. During the testing and trial-and-error stage, the researchers spent approximately ₱2,000 on components such as an ESP32-based RFID reader and other supporting electronic materials. These components were utilized to test the RFID attendance setup and were included in the estimated hardware expenses.

For software, the system utilized development tools and services such as Next.js, MongoDB, Gemini AI, Figma, Visual Studio Code, and GitHub. Most of the development tools used by the researchers were available through free versions or free usage limits. However, database, AI, hosting, and other online services could require additional costs depending on their usage and service plans.

The researchers also utilized existing laptops and personal devices for system development and testing. Since these devices were already available before the project and were not purchased specifically for DC SPACE, they were excluded from the estimated project cost.

The implementation of DC SPACE was projected to generate operational savings by reducing the manual work involved in event registration, RFID attendance monitoring, certificate generation, feedback collection, and report preparation. The system's AI features

also supported event organizers and administrators through attendance prediction, attendance flow analysis, crowd density and congestion prediction, and automated event reports. In addition, the integration with iRoomReserve helped reduce manual work in checking venue availability, capacity, and schedule conflicts.

The following table presented the estimated operational savings:

Category	Monthly Savings	Annual Savings
Time Efficiency	₱1,000	₱12,000
Reduced Manpower	₱500	₱6,000
Error Reduction	₱500	₱6,000
Total Estimated Savings	₱2,000	₱24,000

Table 3.1 Estimated Savings

The system was projected to generate total savings of ₱2,000 monthly or ₱24,000 annually. These savings were based on reduced manual workload, improved processing efficiency, and minimized errors in attendance and certificate generation.

Overall, the relatively low implementation cost combined with the projected operational savings indicated that the system was economically feasible. The expected benefits of automation exceeded the estimated initial investment, supporting the potential long-term sustainability of the project.

3.1.3.3.1 Return of Investment

The Return on Investment (ROI) was computed to determine the projected financial return of the DC SPACE system in relation to its estimated initial investment. It measured the expected gain from the system compared to the amount invested.

Initial Investment	Annual Savings	Net Gain	ROI
₱4,000	₱24,000	₱20,000	500%

Table 3.2 Return on Investment (ROI)

Based on the estimated initial investment of ₱4,000 and projected annual savings of ₱24,000, the system obtained a projected ROI of 500%. This indicated that the expected annual savings were significantly higher than the estimated initial investment.

3.1.3.3.2 Net Present Value

The Net Present Value (NPV) was computed to determine the present value of future cash inflows generated by the system in relation to its initial investment. For this analysis, the annual savings were treated as a recurring cash inflow over a one-year period. A discount rate of 10% was applied to account for the time value of money.

The present value of cash inflows was calculated by dividing the annual savings by (1 + r)^t, where r represented the discount rate and t represented the time period in years. The resulting present value was then compared against the initial investment to determine the net present value.

NPV Computation Variables

Variable	Value
Annual Savings	₱24,000
Discount Rate (r)	10%
Time (t)	1 year

Table 3.3 Net Present Value (NPV) Computation Variables

NPV Result

Present Value of Savings	Initial Investment	NPV
₱21,818.18	₱4,000	₱17,818.18

Table 3.4 Net Present Value (NPV) Result

The resulting NPV was positive at ₱17,818.18, indicating that the project was financially viable. A positive NPV signified that the present value of the expected benefits exceeded the total cost of investment, confirming that the system generated projected economic value even after accounting for the time value of money.

3.1.3.3.3 Payback Period Analysis

The payback period analysis was conducted to determine how long it would take for the system to recover its initial investment based on the projected annual savings generated through its implementation. The computation was based on the total project cost and the estimated annual operational savings derived from time efficiency, reduced manpower requirements, and minimized errors in the attendance process.

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Components	Description	Estimated Cost
Hardware	RFID modules, tags, electronic components	₱4,000
Software	Open-source tools and technologies	₱0
Total Cost		₱4,000

Table 3.5 Payback Period Analysis – Components Description

The total project cost considered in the analysis amounted to ₱4,000, which represented the initial investment required for the development and deployment of the system.

The payback period was computed by comparing the total initial investment with the estimated annual savings generated through system implementation. The total project cost was determined based on the hardware and software expenses and was then divided by the projected annual savings.

Initial Investment	Annual Savings	Payback Period
₱4,000	₱24,000	0.17 years (≈ 2 months)

Table 3.6 Payback Period Computation

Based on the computation, the system was projected to recover its initial investment within approximately two months of operation. This indicated a relatively short recovery period compared to its implementation cost, demonstrating strong economic feasibility in terms of cost recovery and operational efficiency.

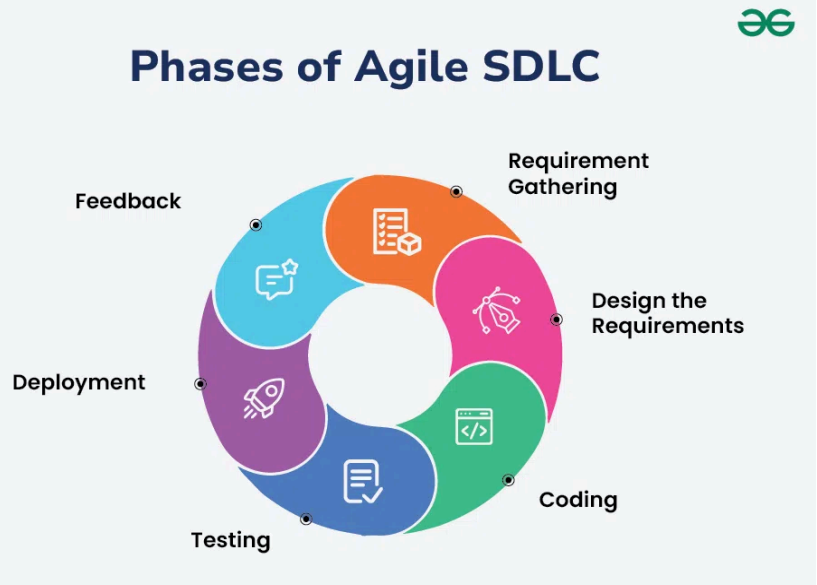
3.1.4 System Design - System Development Life Cycle (SDLC)

The development of DC Space follows the Agile Methodology, an iterative and flexible approach to the Software Development Life Cycle (SDLC). This framework was chosen because it allows the researchers to refine both the hardware (RFID) and software (Next.js and AI) components simultaneously, ensuring the system meets the specific goals of improving efficiency and accuracy at St. Dominic College of Asia.

The process is divided into the following iterative stages based on the study's objectives:

- **Planning:** In this stage, the researchers conceptualized DC Space as a solution for automated attendance and e-certificate issuance. The plan focused on integrating RFID technology with AI-driven analytics to solve manual logging delays and improve the reliability of participation records for both academic and extracurricular activities.
- **Requirement Analysis:** The researchers identified the technical requirements needed to achieve the study's objectives. This included defining the data flow for automated RFID logging, establishing the parameters for the real-time admin dashboard, and determining the data points required for the AI analytics to track active organizations, course engagement, and facility usage.
- **Design:** This phase involved designing the architectural and interface components. The researchers created UI mockups for the user and admin dashboards and mapped out the automated certificate generation system. The design ensures that e-certificates are dynamically produced by merging validated attendance data with institutional templates.

- **Development / Implementation:** During this phase, the actual coding occurred. The team built the web application, including RFID attendance validation, using Next.js, React, and TypeScript. RFID scans are captured through the USB RFID reader and validated against registered accounts directly within the web application. The AI analytics layer was implemented using the Gemini AI API, integrated into the Next.js application to process event trends and support future resource allocation insights.
- **Testing / Integration:** The system underwent rigorous testing to ensure all modules work together. This included verifying that "Tap IN/OUT" data correctly triggers the real-time monitoring dashboard and that the automated delivery system successfully sends e-certificates to registered institutional emails once attendance criteria are met.
- **Maintenance:** In this final stage, the researchers evaluate the system prototype's functionality and reliability. Feedback is used to perform debugging and modifications, ensuring the system remains a usable and efficient tool for improving the certificate issuance and attendance monitoring processes at the institution.



3.1.4.1. User Interface

User Side

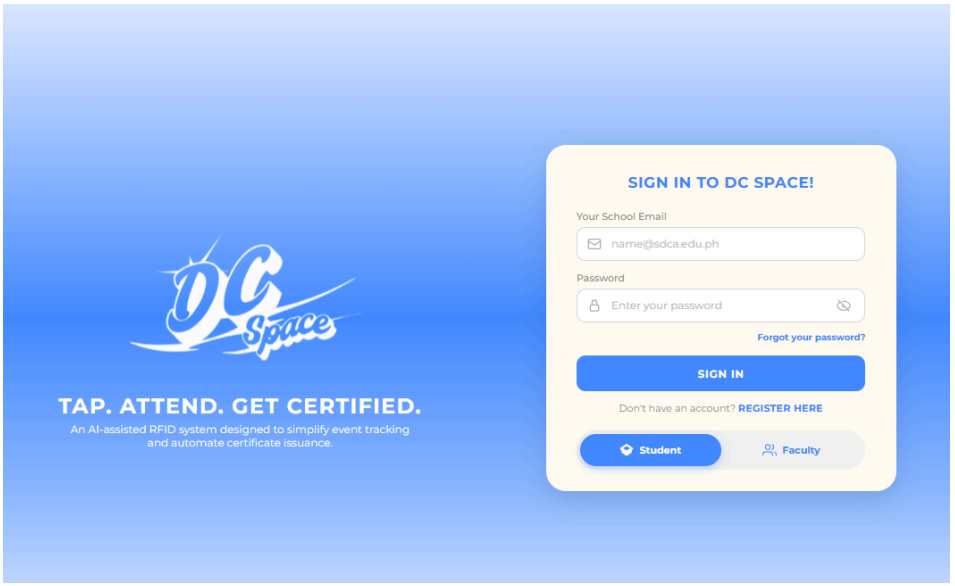


Figure 3.4 User Login Page

The figure presents the Login Page of the DC Space system, which serves as the main entry point for students and faculty members. On the left side of the interface, the DC Space logo, tagline, and a short description of the system are displayed to introduce its purpose. On the right side, a login form is provided where users can sign in using their registered SDCA email address and password. The interface also includes a "Forgot Password" option for account recovery and a registration option for users who do not yet have an account. To register, users are required to provide their SDCA email address and tap their RFID-enabled school ID to verify and create their account. The interface is designed to provide a simple and secure way for students and faculty members to access the system.

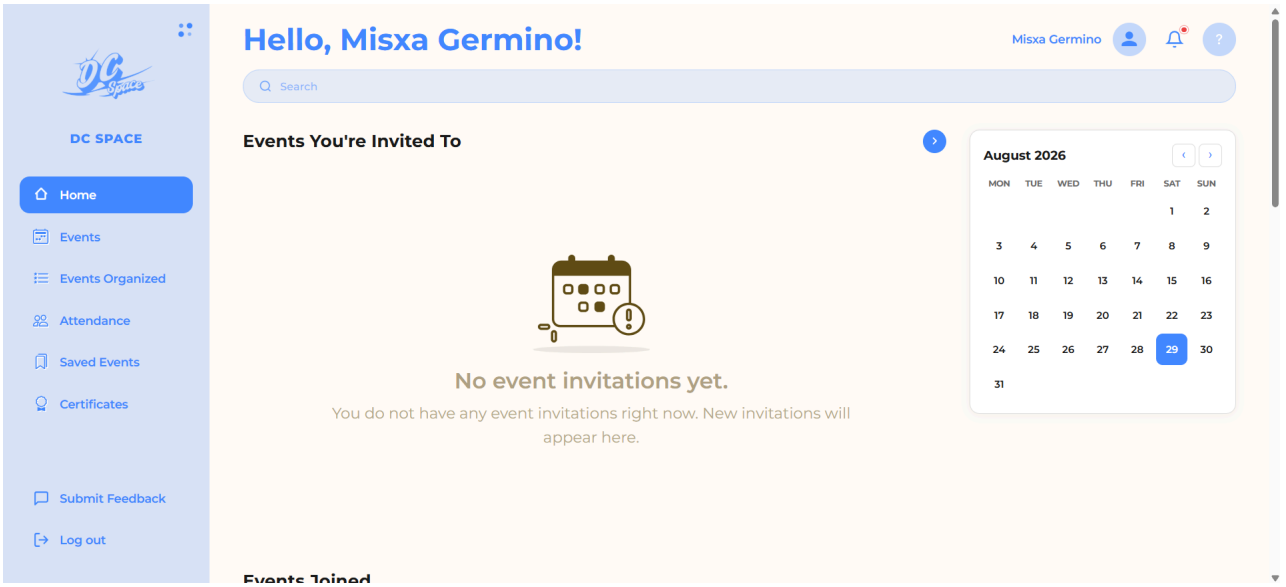


Figure 3.5 User Home Page

The figure presents the Home Page of the DC Space system, which provides users with an overview of their events and activities. It displays the "Events You're Invited To" section for events where the user has been personally invited, along with a calendar showing the scheduled event dates. The page also includes the "Events Joined" section, which displays the user's past, upcoming, and current events. The navigation bar provides access to Events, Events Organized, Attendance, Saved Events, Certificates, and Feedback, while the upper-right section contains the My Profile, Notifications, and Tutorial buttons. The Events Organized option is available only to users who are organization officers or faculty members.

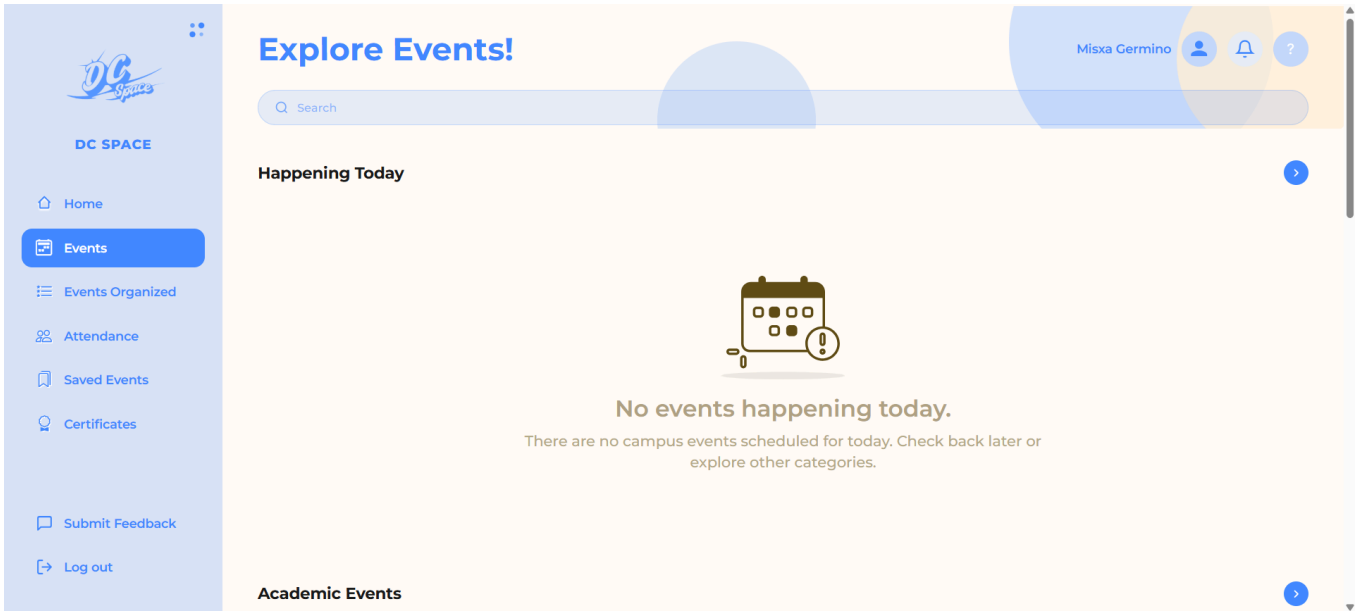


Figure 3.6 User Explore Events Page

The figure presents the Events Page of the DC Space system, which allows students and faculty of St. Dominic College of Asia to discover events they can join. It displays events that are currently happening or scheduled in the future, including events held on and off campus. By selecting an event, users can view important announcements and event details. If an event requires file submissions, users can also submit the required files through the system. The page also notifies users when new events are available.

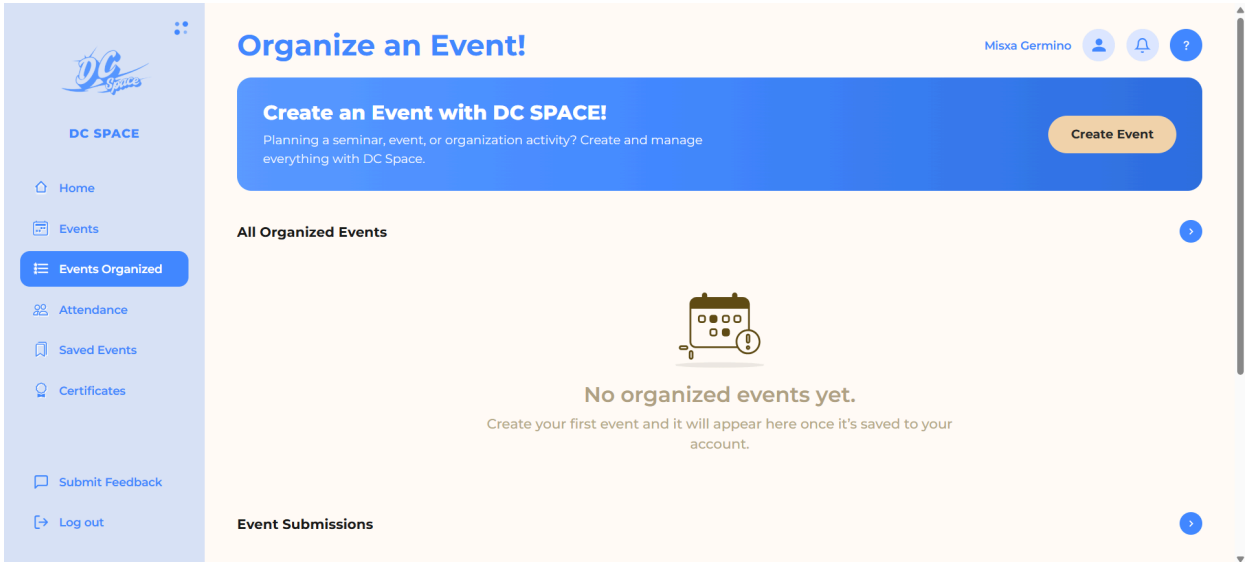


Figure 3.7 User Events Organized Page

The figure presents the Events Organized Page, which is available to organization officers and faculty members who are allowed to organize events. This page allows users to create new events and view events they have organized. It also displays approved events and submitted events based on their status, including pending submissions, saved drafts, and inactive events such as postponed, cancelled, or rejected events. This page provides organizers with a centralized area for managing their event submissions.

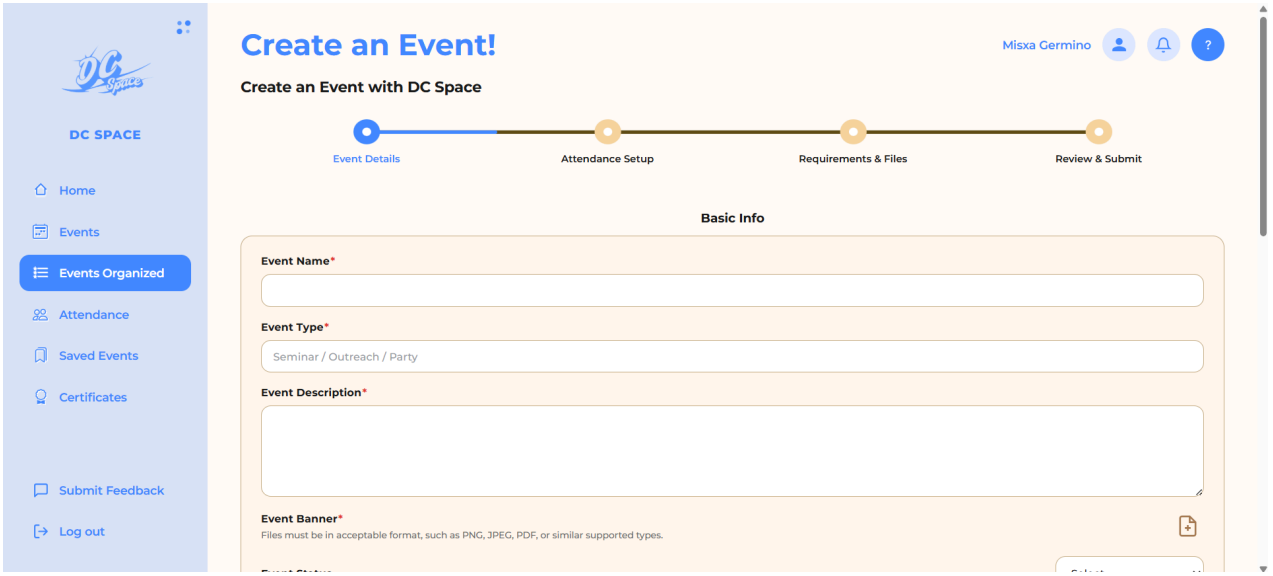


Figure 3.8 User Create Event Form Page

The figure presents the Create Event form, which can be accessed by selecting the "Create Event" button from the Events Organized Page. The form is divided into four steps: Event Details, Attendance Setup, Requirements & Files, and Review & Submit. Each step requires the organizer to provide the necessary information for the event. After completing the form, the organizer can either submit the event for administrator review or save it as a draft for later completion.

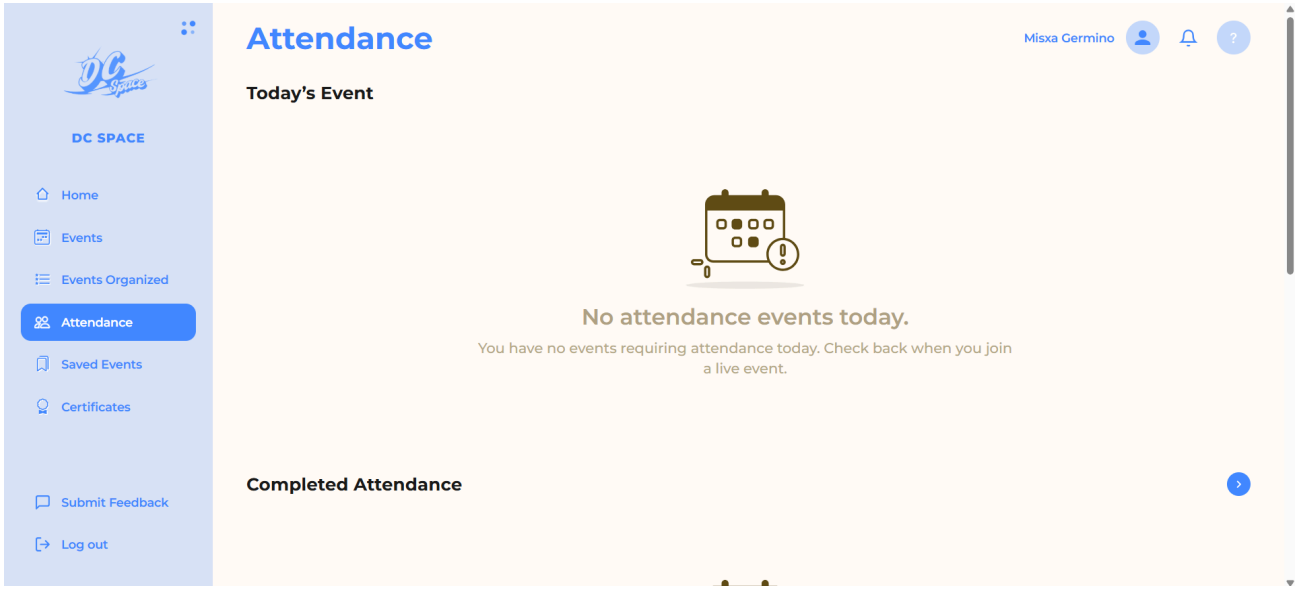


Figure 3.9 User Attendance Page

The figure presents the Attendance Interface of the DC Space system, which allows users to monitor their event attendance records and participation status. This section provides an Attendance Overview, where users can view their attendance status under categories such as Ongoing Attendance, Completed Attendance, and Incomplete Attendance. The interface also includes filter options to help users sort attendance records by date or schedule. This page is designed to help users conveniently track their participation and monitor attendance progress for school events.

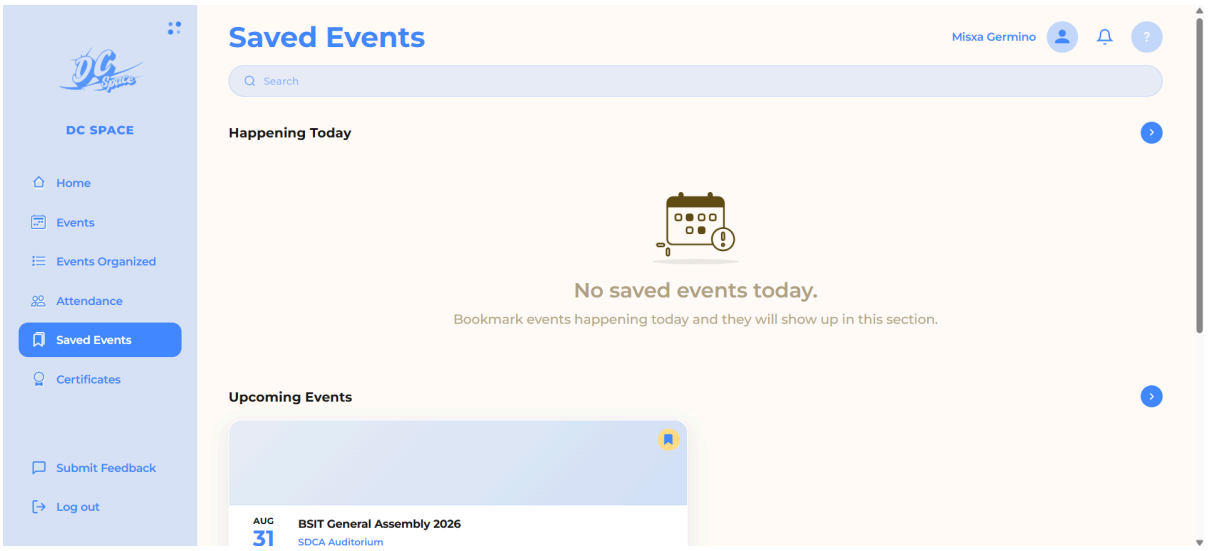


Figure 3.10 User Saved Events Page

The figure presents the Saved Events Interface of the DC Space system, which allows users to view and manage events they have bookmarked for future participation. This section displays the user's saved events, along with search and filter options to help organize events by schedule or preference. If no events have been saved, the system displays a message informing the user and provides a "Browse Events" button for easier access to available school events. This interface is designed to help users conveniently monitor and revisit events they are interested in joining.

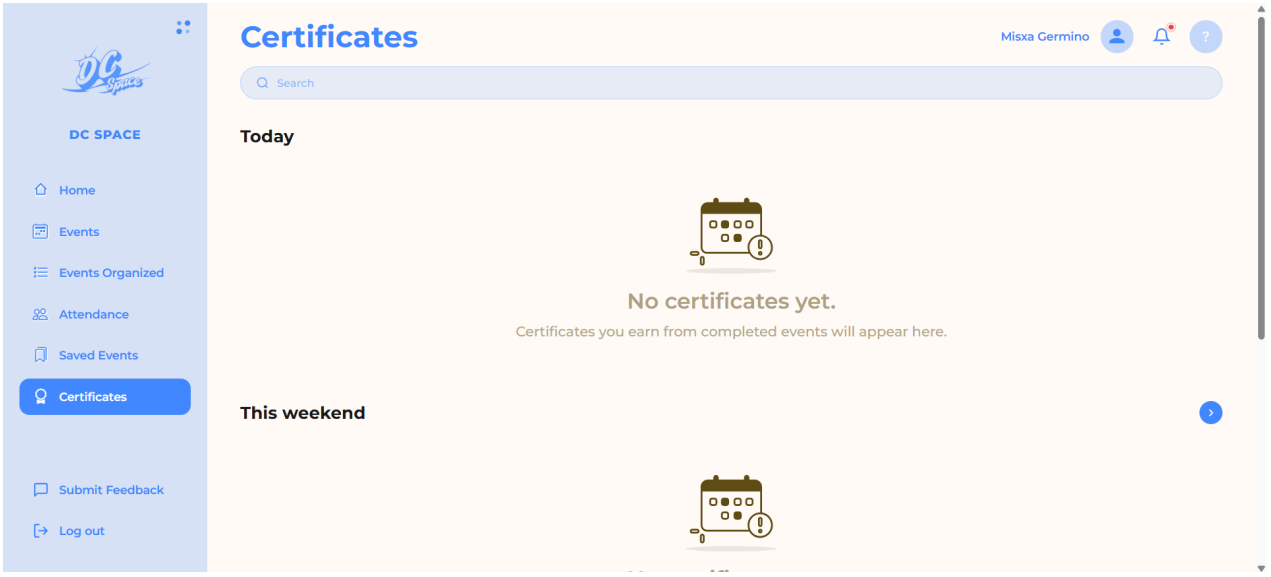


Figure 3.11 User Certificates Page

The figure presents the Certificates Interface of the DC Space system, which allows users to view and access certificates earned from completed school events. This section displays the user's earned certificates in a card-based layout, showing details such as the certificate name, related event, and date issued. The interface also includes filter options to help users organize certificates by date or schedule. This page is designed to provide users with a convenient and organized way to manage and review their earned certificates.

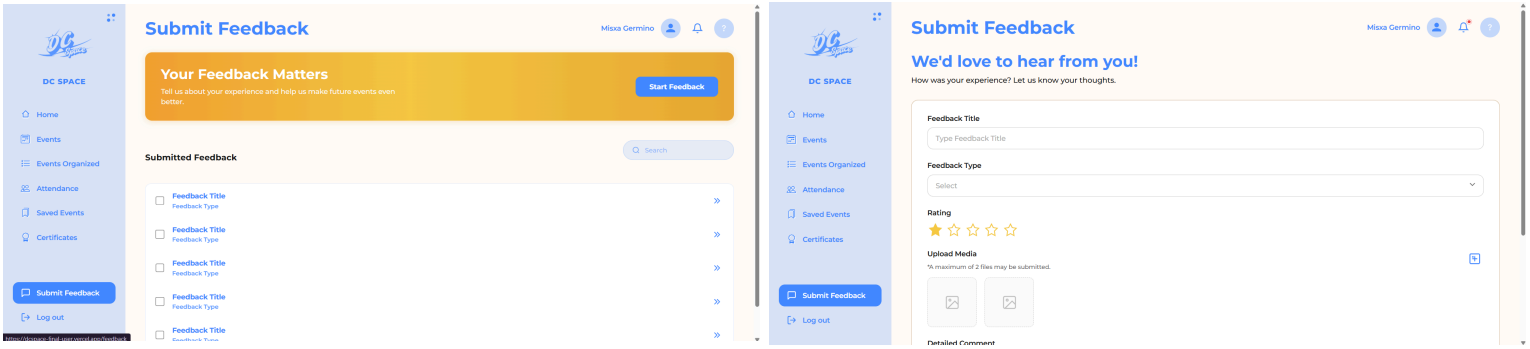


Figure 3.12 User Feedback Page

The figure presents the Feedback Page of the DC Space system, where users can view their previously submitted feedback. It also provides a “Submit Feedback” button that allows users to provide new feedback about the system or an event. The feedback form includes fields for the feedback title and type, star rating, detailed comments, and media uploads. This page provides users with an organized way to submit and review their feedback history.

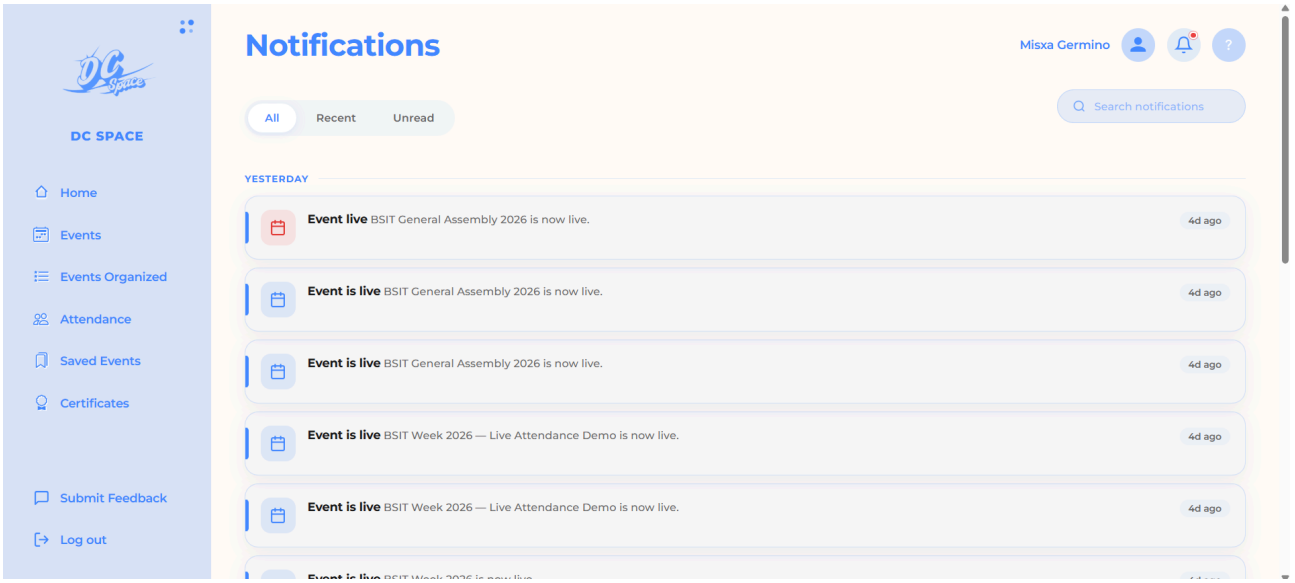


Figure 3.13 User Notifications Page

The figure presents the Notifications Page, which allows users to view and manage important system notifications. It organizes notifications into All, Recent, and Unread sections. Users may receive notifications about new events, event invitations, registration or event errors, attendance completion, and other important system updates. This page allows users to review both recent and previous notifications that they may have missed.

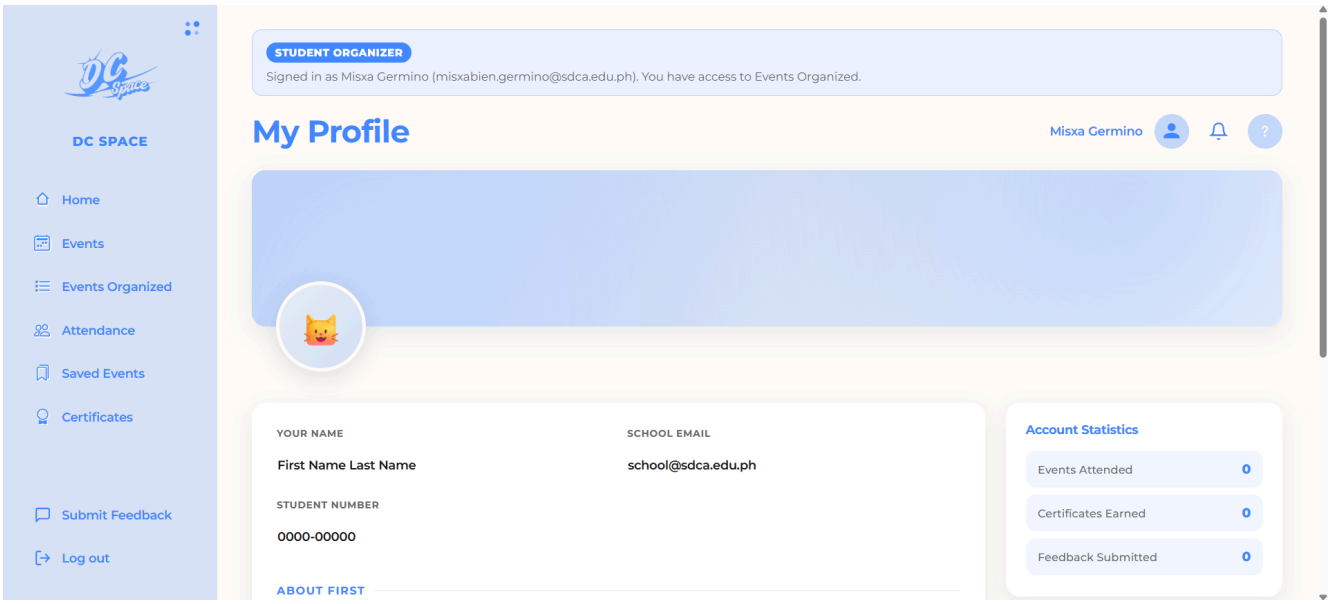


Figure 3.14 User My Profile Page

The figure presents the My Profile Page of the DC Space system, where users can manage and view their personal account information. Users can upload or change their profile and cover photos and view their personal details, account statistics, and recent activities. The account statistics provide a summary of their event participation, certificates earned, and feedback submitted. The page also includes account settings that allow users to change their password when needed.

Administrator and Super Administrator Side

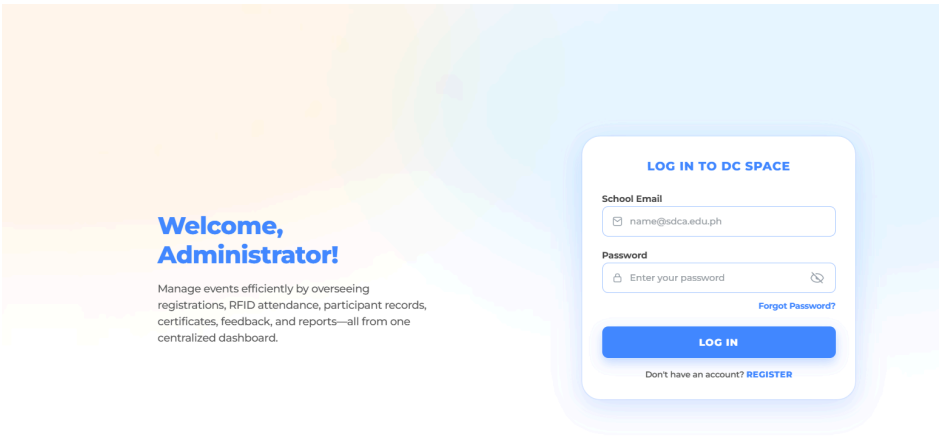


Figure 3.15 Admin and Super Admin Login Page

The figure shows the Sign In page for administrators. Administrators can log in using their official SDCA email account. The page also provides options for changing a password and creating an account. For the super admin, the create account option is not available because a pre-created account is provided for security purposes.

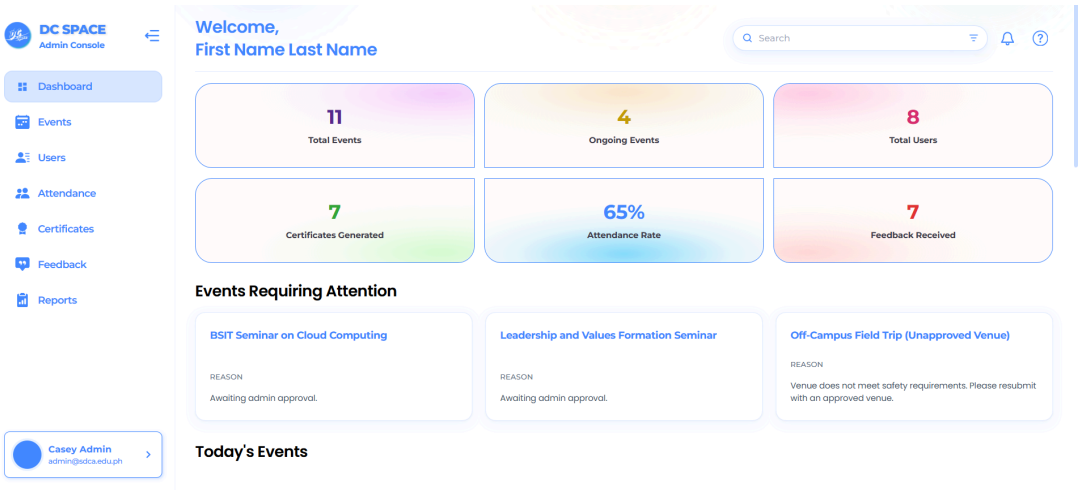


Figure 3.16 Admin and Super Admin Dashboard Page

The figure shows the Dashboard page of the admin and super admin interface. It contains six main statistical boxes showing the total events, ongoing events, total users, certificates generated, attendance rate, and feedback received. It also displays events that require attention, today's events, recent activities, quick charts, notifications, and pending tasks. These sections provide the administrators with a quick overview of the system and current activities.

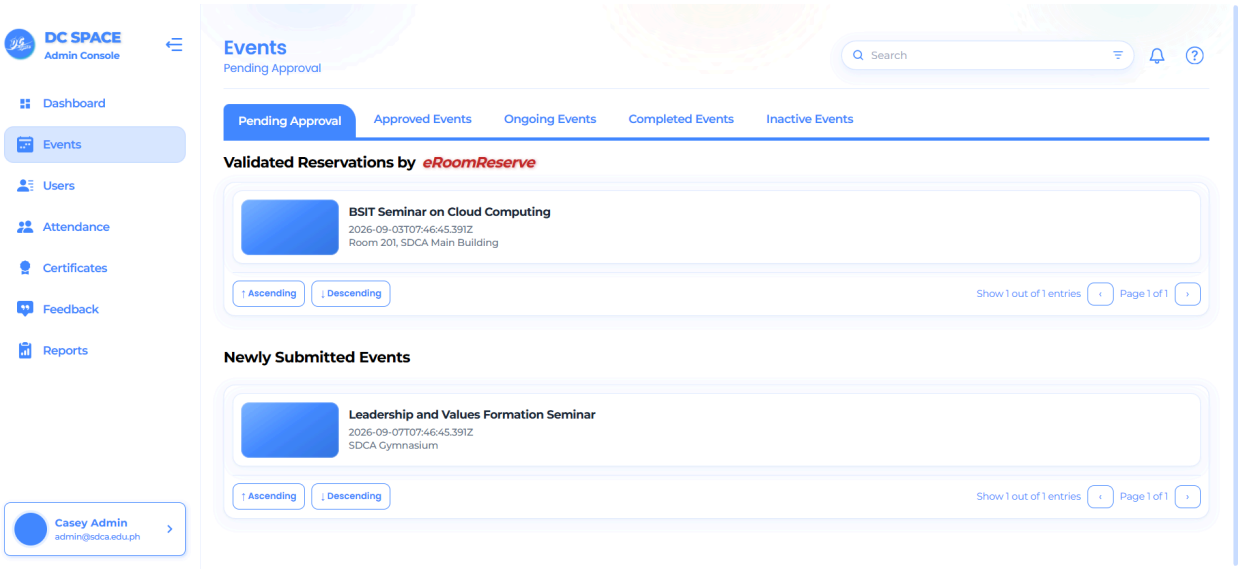


Figure 3.17 Admin and Super Admin Event Management Page

The figure shows the Events page, where administrators can view events based on their status. The Pending Approval section includes events that have already been validated by eRoomReserve and those that are still waiting for validation. The page also provides sections for approved, ongoing, completed, and inactive events. Inactive events include rejected, postponed, and cancelled events.

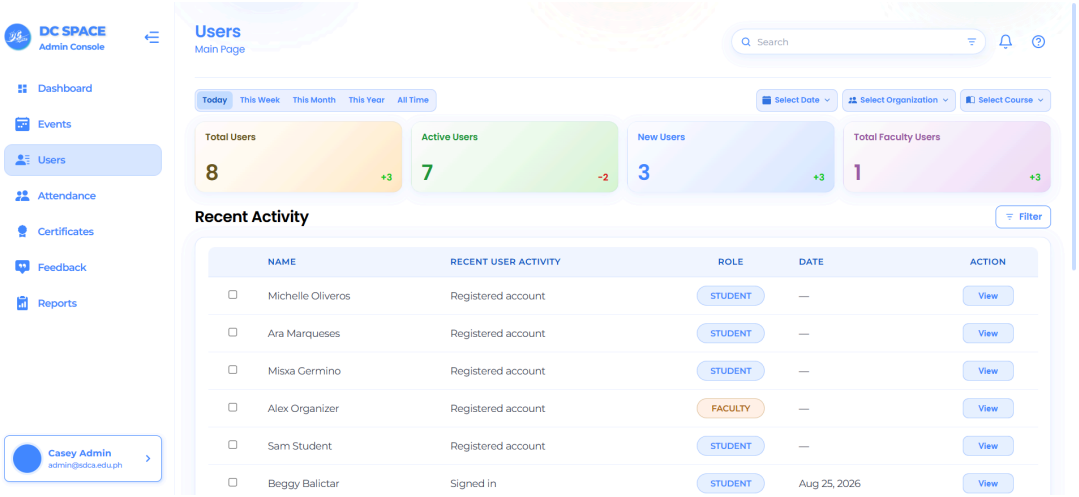


Figure 3.18 Admin and Super Admin User Management Page

The figure presents the main Users page. It contains four statistical boxes showing the total users, active users, new users, and total faculty users. It also displays recent user activities, including the user's name, activity, role, date, and a view option. The page provides options for viewing student and faculty user lists and allows administrators to manually add users.

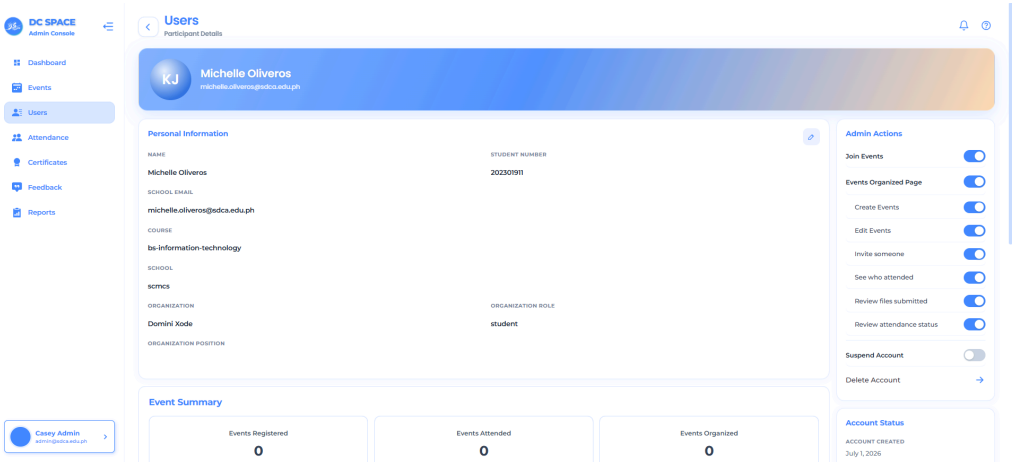


Figure 3.19 User Details Page for Admin User Interface

The figure shows the details of a selected user account. The administrator can view the user's personal information and activities within the system. It also provides system permission controls that allow the administrator to manage the actions available to the user. This page helps administrators monitor user activities and manage their system permissions.

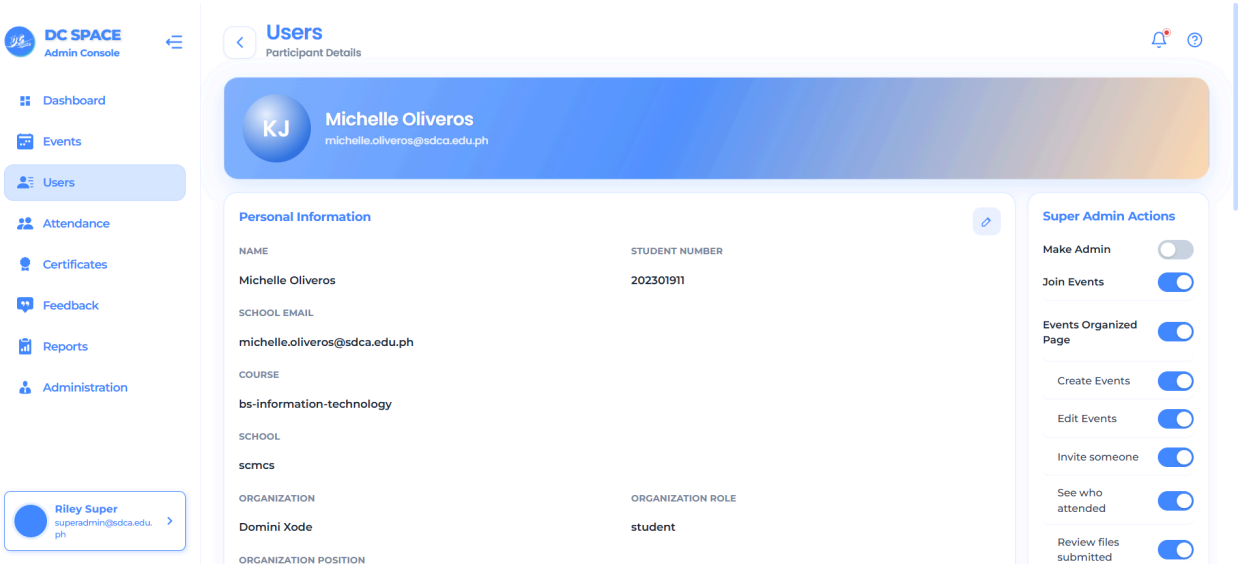


Figure 3.20 User Details Page for Super Admin User Interface

The figure shows the user details page for an account managed by the Super Admin of the DC Space system. The information displayed is similar to the details available to regular administrators, including the user's personal information, account details, organization, and role. However, the Super Admin has additional privileges that allow them to manage administrative access. One of these privileges is the ability to assign a user account as an admin. This provides the Super Admin with greater control over user roles and administrative access within the system.

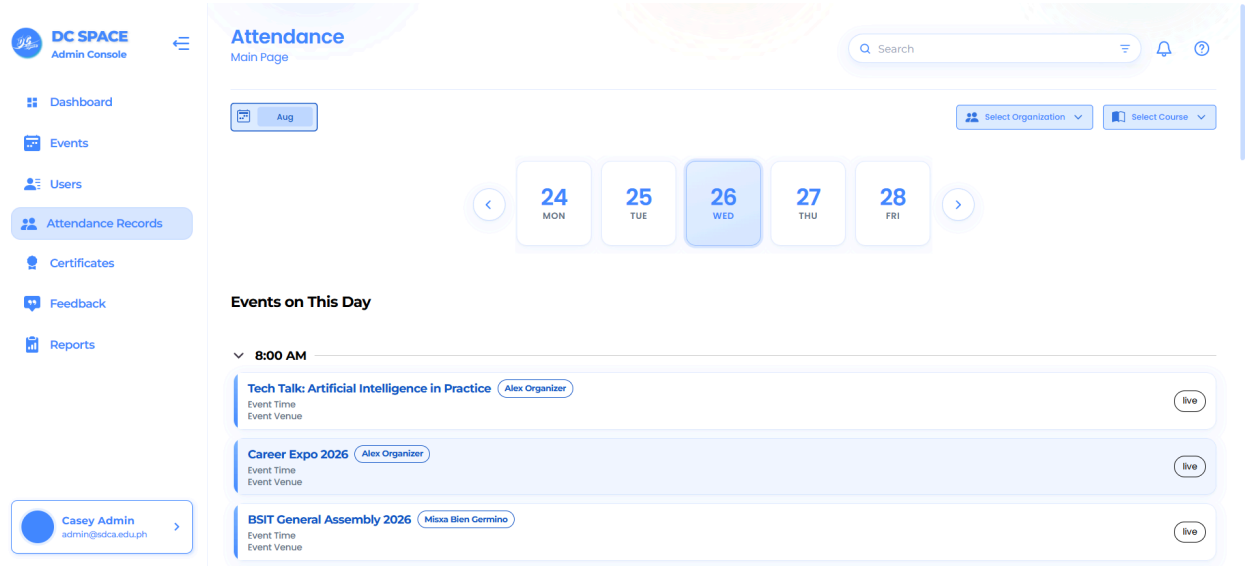


Figure 3.21 Admin and Super Admin Attendance Management Page

The figure shows the Attendance page, where administrators can view past, ongoing, and future events based on the selected date. After selecting an event, the page displays the attendance information for that event, including the attendance results and details of the participants who attended.

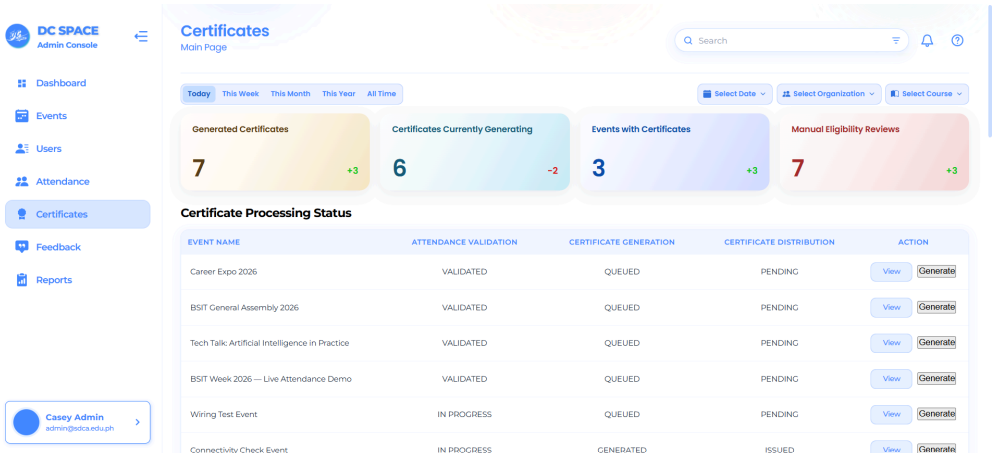


Figure 3.22 Admin and Super Admin Certificates Management Page

The figure presents the Certificates page. It contains four statistical boxes showing the number of generated certificates, certificates currently being generated, events with certificates, and manual eligibility reviews. Below these statistics is the Certificate Processing Status table, which shows the event name, attendance validation, certificate generation, and certificate distribution. The page also includes sections for events requiring attention, processing timelines, and certificate workflow progress. This allows administrators to monitor and manage the certificate generation process.

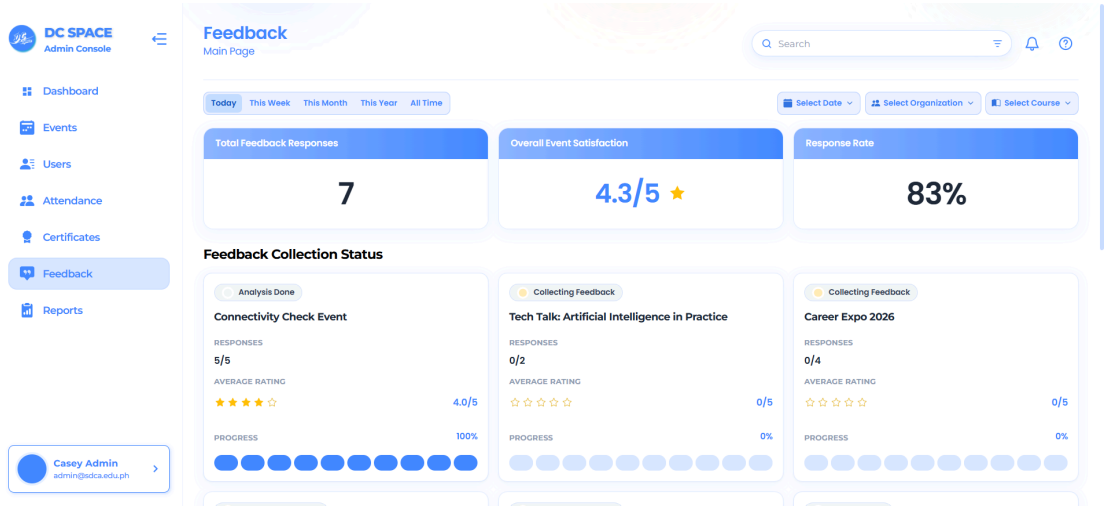


Figure 3.23 Admin Feedback Management Page

The figure shows the Feedback page, which contains three main statistical boxes for total feedback responses, including mid-event feedback, post-event feedback, system issues, and other feedback. It also displays the overall event satisfaction and response rate. Administrators can view the feedback collection status, while super administrators can also access system feedback submitted by users. The page includes category ratings that show the overall star rating for areas such as registration, events, and the system.

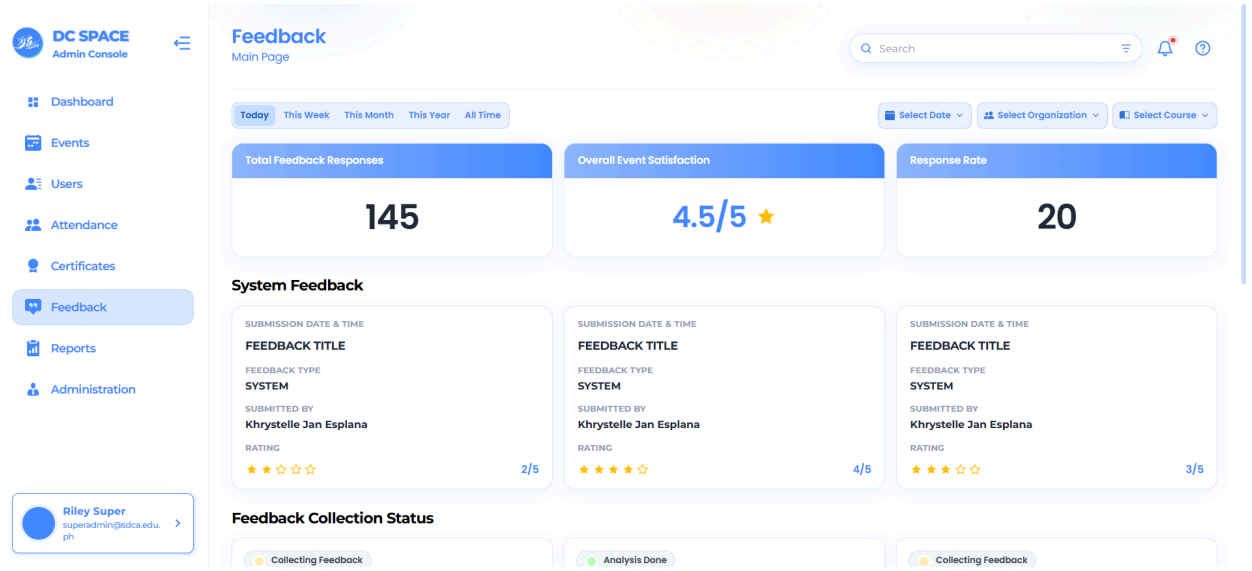


Figure 3.24 Super Admin Feedback Management Page

The figure shows the main Feedback page when using a Super Admin account. The Super Admin can access the same feedback functions available to regular administrators, but has an additional section for viewing system-related feedback submitted by users. These feedback entries may include concerns or issues related to the system that require attention. To help monitor these concerns, the page includes feedback statuses such as “Resolved,” “Reviewing,” and “Archived.” This allows the Super Admin to track the progress of reported system issues and determine whether they have already been addressed or still require further action.

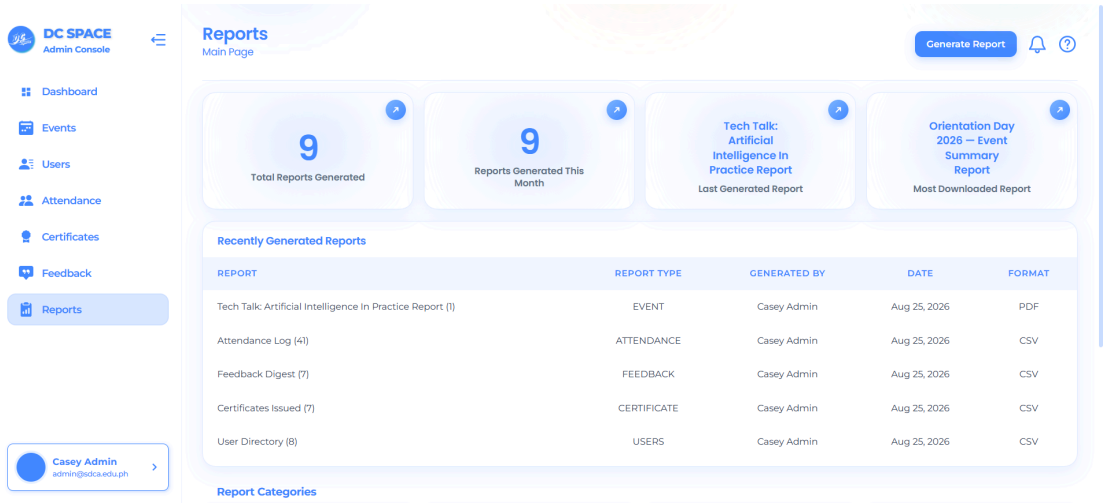


Figure 3.25 Admin and Super Admin Report Management Page

The figure presents the Reports page. It contains four statistical boxes showing the total reports generated, reports generated within the current month, the most recently generated report, and the most downloaded report. It also displays recently generated reports and different report categories, including event, attendance, certificate, feedback, and user directory reports. The page allows administrators to manage existing reports and generate new reports with the help of AI.

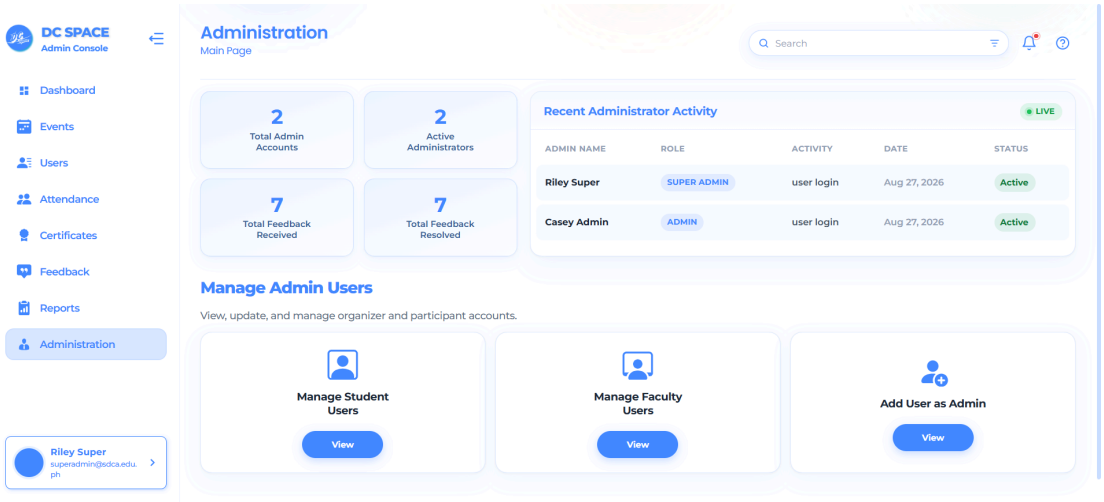


Figure 3.26 Super Admin Administration Management Page

The figure shows the main Administration page, which is exclusively accessible to the Super Admin. This page is designed to manage and monitor administrator accounts within the DC Space system. It contains four statistical summary boxes showing the Total Admin Accounts, Active Administrators, Total Feedback Received, and Total Feedback Resolved. The Super Admin can also manage the system permissions of administrator accounts to control the functions and level of access available to each admin. In addition, the page provides access to audit logs for each administrator account. These logs allow the Super Admin to monitor the activities performed by administrators and keep track of important actions within the system.

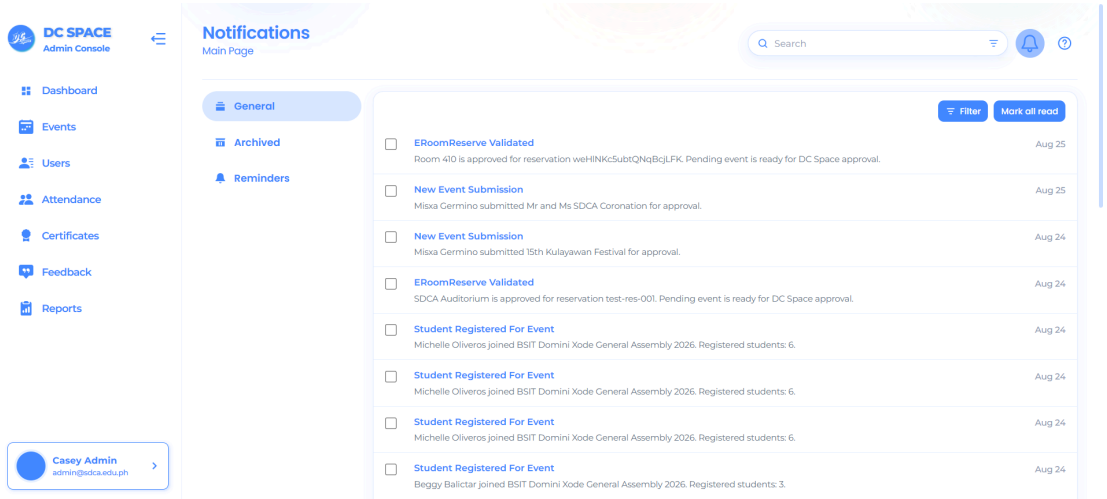


Figure 3.27 Admin and Super Admin Notifications Page

The figure shows the Notifications page, which contains general notifications, archived notifications, and reminders. The page informs administrators and super administrators about important system activities, such as newly registered or eRoomReserve-validated events, system errors or issues, and events that require monitoring or management.

3.1.4.2. System Input and Output

This section presents the data entered into the DC SPACE system and the information produced after the system processes these data. It shows how user actions, RFID scans, event information, feedback, and AI-related data are used to support event management, attendance monitoring, and event evaluation.

System Input

These are the data points the system collects from users, hardware, and external services to function:

- **RFID UID (Unique Identifier):** The unique ID read from the student's or faculty member's RFID-enabled institutional ID card during tap-in and tap-out.
- **User Information**: Personal and academic information entered during account registration, such as name, school email, student or faculty number, course, section, and organization.
- **Login Information:** The school email and password entered by users when accessing their accounts.
- **Event Information:** Details entered by event organizers, such as event title, description, venue, date, time, attendance requirement, and required files.
- **Registration Data:** Information submitted when users register for an event, including required files when needed.
- **Attendance Data:** RFID tap-in and tap-out records, including the participant ID, event, and timestamps.

- **Feedback and Survey Data:** Ratings, comments, and survey responses submitted by participants during or after an event.
- **Event Media:** Photos uploaded by authorized users or event organizers for event records and archiving.
- **Administrative Data:** Information and actions entered by administrators, such as event approvals, user management, attendance checking, certificate processing, and feedback monitoring.
- **AI Analysis Data:** Event and attendance data used by the AI module to generate attendance predictions, crowd and congestion insights, and event recommendations.
- **Venue Information:** Venue and scheduling information used to validate event reservations and capacity through the iRoomReserve integration.

System Output

These are the results and notifications generated after the system processes the inputs:

- **Attendance Records:** Real-time tap-in and tap-out logs showing the participant's attendance time, duration, and attendance status.
- **Event Information and Status:** Event listings and updates showing events that are upcoming, ongoing, completed, pending, rejected, cancelled, or other applicable statuses.
- **AI Event Insights:** AI-generated results such as expected attendee counts, attendance trends, crowd density, possible congestion, and recommendations for event organizers.

- **Attendance Security Results:** Alerts or records for duplicate taps, invalid scans, and other suspicious attendance activities.
- **Automated E-Certificates:** Digital certificates generated for participants who meet the required attendance conditions.
- **Event Reports:** Generated reports containing attendance data, participant feedback, event results, AI insights, recommendations, and other event information.
- **Feedback and Survey Results:** Collected ratings, comments, and survey responses that can be reviewed by authorized administrators.
- **Event Media Archive:** Stored event photos and other approved media that can be viewed as part of the event records.
- **Notifications:** System notifications for event updates, registration, attendance, certificates, feedback, approvals, and other important activities.
- **Administrative Records:** User, event, attendance, certificate, feedback, and system activity records that can be viewed and managed by authorized administrators.
- **System and Audit Logs:** Records of important user and administrative actions for monitoring and system management.

3.2. Project Development Outline

3.2.1. Gantt Chart

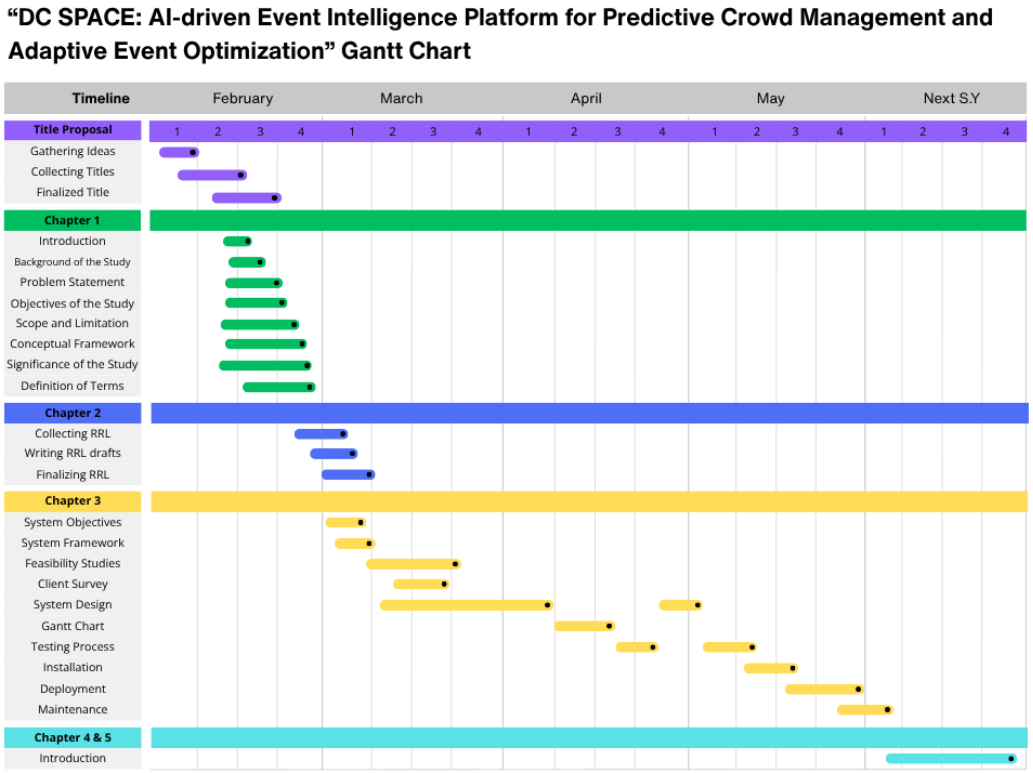


Figure 3.6 Gantt Chart Table

The Gantt chart presented the timeline and progress of the development of DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. The project started with the title proposal phase in February, followed by the completion of Chapters 1 and 2. Chapter 3 required the longest development period because it involved important activities such as system design, testing, installation, deployment, and maintenance. This phase also involved numerous trial-and-error activities to improve the system's functionality and enhance the website's design to make it more appealing and user-friendly. The researchers dedicated additional time to testing and revising

the system to ensure that it functioned properly and addressed the needs of its intended users. Lastly, Chapters 4 and 5 were scheduled for the following school year to present, interpret, and analyze the gathered data and project results.

3.2.2. Testing Process

The researchers conducted different tests to evaluate the functionality, reliability, and usability of the developed DC SPACE system. The testing focused on the system's major features, including user account management, event management, RFID attendance, certificate generation, database operations, and user interface.

The testing process also helped identify errors, technical issues, and areas that required improvement. The identified problems were addressed through necessary adjustments and revisions to ensure that the system performed according to its intended functions before deployment.

3.2.2.1. Functional testing

Test Case	**Objective**	**Expected Result**	**Actual Result**	**Status**
User Registration	To verify if users can successfully create an	Users should be able to register their account	The system successfully created user accounts after	Passed

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	account in the system	successfully	the required information was provided	
RFID Tap In and Tap Out	To test if the RFID attendance process records attendance correctly	The system should record both tap in and tap out time accurately	The tap in and tap out attendance logs were recorded correctly in the system	Passed
Dashboard updates	To check if event and attendance information is updated	The dashboard should display updated event and attendance information	The dashboard successfully displayed the user's event and attendance information	Passed
Event Registration	To verify if users can register for an event	The user should be registered and the event should appear in their event	The user can register for the event and show it in their 'Events Joined' lists.	Passed

		records		
Certificate Generation	To check if certificates are generated for qualified participants	The system should generate a certificate based on the attendance record	The system has successfully generated the certificate once the attendance requirement was met.	Passed
Notification	To verify if users receive system notifications.	The system should display the appropriate notification.	The system has displayed appropriate notifications for the user.	Passed

Table 3.7 Functional Testing Table

Functional testing was conducted to verify whether the main features of DC SPACE operated according to their intended functions. The researchers tested user registration, RFID tap-in and tap-out, dashboard updates, event registration, certificate generation, and notifications. The results showed that the system successfully created user accounts after the required information was provided. RFID tap-in and tap-out attendance logs were also recorded correctly. The dashboard displayed updated event and attendance information, while

event registration successfully added users to their respective joined events. Furthermore, the system successfully generated certificates once participants met the required attendance conditions and displayed appropriate notifications to users. All functional testing cases received a Passed status, indicating that the tested features performed according to their expected results.

3.2.2.2. User Interface Testing

Test Case	**Issue Found**	**Solution Implemented**	**Status**
Mobile Responsiveness	Navigation texts and buttons overlapped on smaller screen sizes	The responsive layout and spacing of the interface were adjusted and improved	Fixed
Error Messages	Error messages did not appear when users entered invalid or incomplete school email addresses	Validation and error message prompts were added to the sign up and login forms	Fixed
Loading Screens	No loading screen	A customized	Fixed

	was displayed while pages were loading, causing delays in user feedback	animated loading screen was designed and implemented	
Navigation	Some page navigation required improvement during testing	Navigation and page links were revised	Fixed
Form Validation	Some forms required additional validation	Additional validation was applied to required fields	Fixed

Table 3.8 User Interface Testing Table

User interface testing was conducted to identify issues related to the system's appearance, responsiveness, navigation, and overall usability. The researchers identified several issues during testing, including overlapping navigation texts and buttons on smaller screen sizes, missing error messages when users entered invalid or incomplete school email addresses, the absence of loading screens during page transitions, and navigation-related problems. To address these issues, the researchers improved the responsive layout and spacing of the interface, added validation and error message prompts to the sign-up and login

forms, implemented a customized animated loading screen, and revised the navigation elements. The identified user interface issues were addressed through the necessary revisions and improvements, resulting in a more responsive and user-friendly system.

3.2.2.3. Database Testing

Test Case	Expected Result	Actual Result	Status
Attendance logs	RFID attendance records should be stored accurately	Attendance records and timestamps were successfully stored in MongoDB	Passed
Tap in/out times recorded	Attendance times should be displayed correctly	The attendance page displayed the correct tap-in and tap-out records	Passed
Event records	Event information and participation records should be saved correctly	Event participation records were successfully updated	Passed

User Records	User information should be stored and retrieved correctly	The user information is stored and retrieves complete information correctly in the system.	Passed
Certificate Records	Generated certificate information should be stored correctly	The certificate data was saved correctly, however improvements to the certificate's visual design were identified separately.	Passed

Table 3.9 Database Testing Table

Database testing was conducted to verify whether the system properly stored and retrieved information from the MongoDB database. The researchers tested attendance logs, tap-in and tap-out records, event records, user records, and certificate records. The results showed that RFID attendance records and their corresponding timestamps were successfully stored in MongoDB. The attendance page also displayed the correct tap-in and tap-out information. Event participation records were successfully updated, while user information was properly stored and retrieved. Generated certificate information was also saved correctly, although improvements to the certificate's visual design were identified separately. The

results verified that the system's database operations functioned properly in storing and retrieving the tested records.

3.2.2.4. RFID Testing

Test Case	Expected Result	Actual Result	Status
Tap response time	The RFID reader should immediately detect and scan school IDs	The RFID reader successfully scanned IDs with minimal delay during testing	Passed
Multiple tap testing	The RFID reader should handle consecutive RFID taps properly	The system successfully processed multiple RFID taps without crashing or duplicating records	Passed
Current student IDs can be scanned on the RFID reader	School IDs should be recognized and scanned by the RFID reader.	The RFID reader successfully detected and scanned current	Passed

		student IDs during testing	

Table 3.10 RFID Testing Table

RFID testing was conducted to evaluate the performance and responsiveness of the RFID reader used in the system. The researchers tested the tap response time, multiple consecutive RFID taps, and the scanning of current student identification cards. The results showed that the RFID reader successfully detected and scanned school IDs with minimal delay during testing. The system also handled multiple consecutive RFID taps without crashing or generating duplicate attendance records. Furthermore, the RFID reader successfully recognized and scanned current student IDs during the testing procedures. All RFID testing cases received a Passed status, confirming that the RFID reader functioned properly during testing.

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3.2.2.5. Role and Access Testing

Test Case	Expected Result	Actual Result	Status
Student/Faculty Access	Students and faculty should only access features available to regular users	Students and faculty can access their available features, such as browsing and registering for events, viewing attendance, certificates, and submitting surveys and feedback	Passed
Event Organizer Access	Event organizers should be able to manage their events	Event organizers can manage their events, including creating events and viewing their pending approvals, drafts, rejected events, and other event records. |	Passed
Administrator	Administrators	Administrators are	Passed

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Access	should access administrative functions allowed by their permissions	able to access appropriate administrative functions.	
Super Administrator Access	Super administrators should access administration and system management functions	Super Administrators can access administration features, manage Admin accounts, view system-related feedback, and manage system permissions.	Passed
Admin Permission Control	Admin permissions should follow the settings assigned by the Super Administrator	Admin can follow necessary settings that are assigned only by the Super Administrator.	
Super Admin Permission Control	Super Administrators should be able to	Super Administrators can view an Admin	Passed

	assign or limit the system permissions of Admin accounts	account and manage the system permissions assigned to the Admin, which controls the functions the Admin can access	

Table 3.11 Role and Access Testing Table

Role and access testing was conducted to verify the different access levels and permissions assigned to users within DC SPACE. The researchers tested the access of students and faculty members, event organizers, administrators, and super administrators. The testing also verified the Super Administrator's ability to manage administrator accounts and assign system permissions. The results showed that students and faculty members were able to access the functions assigned to regular users, including event browsing, registration, attendance monitoring, certificate access, surveys, and feedback submission. Event organizers were able to manage their events, including creating events and viewing pending approvals, drafts, rejected events, and other event records. Administrators were able to access the administrative functions allowed by their assigned permissions. Meanwhile, Super Administrators were able to manage administrator accounts, assign system permissions, view system-related feedback, and access system management features. The results confirmed that users were able to access the functions assigned to their respective roles and that the role-based access control features worked according to their intended permissions.

3.2.2.7. AI Event Intelligence Testing

Test Case	Expected Result	Actual Result	Status
AI Attendance Flow Analysis	The system should analyze attendance records and provide information about the flow of participants during the event	The system successfully analyzed the available attendance records and provided attendance flow information. Minor adjustments were needed to improve the presentation and accuracy of the generated analysis.	Passed with improvements
Predictive Expected Attendees	The system should provide an estimated number of expected attendees based on available event and registration data	The system successfully generated an estimated number of expected attendees using the available event and	Passed

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		registration data.	
AI Crowd Density Analysis	The system should provide an analysis of crowd density based on available attendance data	The system successfully provided crowd density information based on the available attendance data and bookmarked events to registered event data.	Passed
Congestion Prediction	The system should identify possible crowd congestion based on available event and attendance information	The system was able to provide congestion predictions, but some results required further adjustment to improve the accuracy of the generated prediction.	Passed with improvements

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Duplicate Entry Detection	The system should identify possible duplicate or repeated attendance entries	The system was able to identify repeated attendance scans, but additional conditions were needed to reduce possible false detections.	Passed with improvements
Suspicious Scanning Detection	The system should identify unusual or suspicious RFID scanning behavior	The system was able to identify possible suspicious scanning patterns, but the detection conditions required further adjustment for more accurate results.	Passed with improvements
AI-Generated Report	The system should generate an event report based on available event, attendance, and survey data	The system successfully generated an event report using the available event information,	Passed

		attendance records, and survey results.	
AI-Generated Program Flow	The system should generate a suggested program flow based on the provided event information	The system successfully generated a suggested program flow based on the event information provided by the organizer.	Passed

Table 3.12 The AI Event Intelligence Testing Table

AI Event Intelligence testing was conducted to evaluate the performance of the system's AI-powered features. The researchers tested the system's ability to analyze attendance flow, predict expected attendees, analyze crowd density, predict congestion, detect duplicate and suspicious scans, and generate event reports and program flow. The results showed that the system successfully analyzed available attendance records and provided attendance flow information, although minor adjustments were needed to improve the presentation and accuracy of the generated analysis. The system also successfully generated estimated expected attendee counts using available event and registration data. Crowd density information was generated based on available attendance data and bookmarked events in relation to registered event data. For congestion prediction, the system was able to provide

predictions based on event and attendance information. However, some results required further adjustments to improve their accuracy. The duplicate entry detection feature successfully identified repeated attendance scans, but additional conditions were needed to reduce possible false detections. Similarly, the suspicious scanning detection feature identified possible suspicious scanning patterns, although its detection conditions required further refinement. Lastly, the system successfully generated event reports using available event information, attendance records, and survey results. It also successfully generated suggested program flows based on the event information provided by organizers. Overall, most AI Event Intelligence features produced the expected results during testing. Minor improvements were identified in attendance flow analysis, congestion prediction, duplicate entry detection, and suspicious scanning detection. The identified issues were addressed through further adjustments to the system's conditions and analysis processes to improve the accuracy and reliability of the generated results.

3.2.2.7. Problems Encountered during testing

Problem Encountered	**Solution Applied**
Invalid school emails were accepted	Added email validation for @sdca.edu.ph accounts
Navbar layout overlapped on smaller devices	Revised responsive CSS layout
Slow loading feedback during page transitions	Added loading screen animation

Certificate layout required further improvement	Revised the certificate layout and visual elements while keeping the certificate data storage and generation functions intact
AI attendance flow analysis needed improvement in presenting and interpreting some attendance patterns	Adjusted the analysis process and output presentation
Congestion predictions required further adjustment for more accurate results	Improved the conditions used for generating congestion predictions
Duplicate attendance scans could result in possible false detections	Added additional conditions when identifying duplicate entries
Suspicious scanning detection required further adjustment to identify unusual patterns more accurately	Refined the conditions used to identify suspicious scanning behavior

Table 3.13 Problems Encountered during Testing Table

During the testing process, the researchers encountered several technical and interface-related issues that required further improvements. The first issue involved the acceptance of invalid school email addresses. To address this, the researchers implemented email validation to ensure that only accounts using the required @sdca.edu.ph email format were accepted. Another issue involved the navigation bar layout, which overlapped on smaller devices. The researchers revised the responsive CSS layout to improve the

appearance and accessibility of the navigation elements. The researchers also encountered slow loading feedback during page transitions. To improve the user experience, a loading screen animation was implemented to provide visual feedback while pages were loading. Additionally, the certificate layout required further improvement. The researchers revised the certificate's layout and visual elements while maintaining its existing data storage and generation functions. Several AI-related issues were also identified during testing. Attendance flow analysis required improvements in presenting and interpreting attendance patterns, while congestion predictions needed further adjustments to generate more accurate results. Furthermore, duplicate attendance scans could result in possible false detections. Additional conditions were therefore implemented to improve duplicate entry identification. The suspicious scanning detection feature also required refinement to identify unusual scanning patterns more accurately. Through these adjustments, the researchers improved the system's accuracy, usability, responsiveness, and overall performance.

3.2.3. Installation and Deployment

The installation and deployment of the DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization will be conducted within St. Dominic College of Asia (SDCA) for school event management and attendance monitoring. The system will be deployed as a web-based application that can be accessed by students, faculty members, event organizers, administrators, and Super Administrators using devices connected to the internet.

During installation, the system will be configured using the required software, database, and external services to support its main functions. The web application, developed using Next.js, will be connected to MongoDB for storing user information, event details, attendance records, certificates, feedback, and other system data. The Google Gemini API will also be configured to support the AI-powered event analytics and reporting features. The system will also be configured to support the iRoomReserve integration for checking venue reservation, schedule, and capacity information.

For deployment, a USB RFID reader will be placed at designated event entrances or attendance areas where students and faculty members can use their RFID-enabled institutional IDs for Tap-In and Tap-Out. Users will be able to access the web application to browse events, register for activities, view attendance records, receive notifications, submit feedback, and access their earned certificates. Event organizers will be able to create and manage events, while administrators and Super Administrators will be able to monitor events, attendance, feedback, reports, and other system activities based on their assigned permissions.

The AI features will also be used to provide event insights, such as expected attendance, attendance trends, crowd density, possible congestion, and other recommendations. After an event, the system will generate event reports using the available attendance, feedback, and event data. These features will help organizers and administrators make better decisions before, during, and after school events.

3.2.4. Maintenance

The maintenance of the DC SPACE system is important to ensure that the system continues to work properly during school events at St. Dominic College of Asia. Regular maintenance will include checking the database to ensure that user information, event details, attendance records, certificates, feedback, and other system data are properly stored and updated.

The USB RFID reader will also be checked regularly to ensure that institutional IDs are properly detected during Tap-In and Tap-Out. If problems such as failed scans or device errors occur, hardware checking and troubleshooting will be performed.

Software maintenance will include fixing bugs, improving system performance, updating system features, and addressing issues reported by users, event organizers, administrators, and Super Administrators. The AI features and external services will also be monitored to ensure that they continue to provide proper results. Database backups and regular system checks may be performed to help prevent data loss.

System permissions and user accounts will also be reviewed to ensure that users only have access to the features allowed for their assigned role. Updates and improvements may be applied based on user feedback and system evaluation results to maintain the usability, reliability, and overall performance of DC SPACE.

CHAPTER 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents, analyzes, and interprets the data gathered from the evaluation of DC SPACE: AI-driven Event Intelligence Platform for Predictive Crowd Management and Adaptive Event Optimization. The evaluation determined the system's quality based on eight software quality characteristics. Responses were collected from end-users and IT experts and analyzed using the mean, standard deviation, correlation, and Cronbach's alpha.

4.1 Sampling Design

The researchers used purposive sampling in selecting respondents for the evaluation of DC SPACE. The respondents were divided into two groups: end-users and IT experts. End-users were individuals who represented the intended users of the platform within St. Dominic College of Asia, including students and other individuals involved in academic and extracurricular events. Their evaluation focused on the functions and user experience of the platform, such as event browsing and registration, RFID attendance, certificates, feedback, and other user-related functions.

The IT experts were selected based on their knowledge or experience in information technology, software development, systems analysis, system administration, or related fields. Their evaluation focused on the technical and software quality aspects of DC SPACE.

The researchers obtained two perspectives in evaluating the system by including both end-users and IT experts. End-users provided feedback based on their use of the system, while IT experts provided a technical assessment of the system's implementation and software quality.

4.1.1 Respondent Distribution

The study planned to include a total of 100 respondents, consisting of 80 end-users and 20 IT experts. The 80 end-users were planned to be distributed equally among five selected SDCA departments, with 16 respondents from each department. The remaining 20 respondents were IT experts who met the criteria established by the researchers.

Table 4.1: Distribution of Respondents

Respondent Group	Number of Respondents	Percentage
End-Users	80	80%
IT Experts	20	20%
Total	100	100%

The table above shows the planned distribution of the respondents. The end-users represented 80% of the total respondents, while the IT experts represented 20%. This distribution provided the study with both user-based and technical perspectives in evaluating DC SPACE.

For the end-user group, the researchers planned to obtain 16 respondents from each of the five selected SDCA departments, resulting in 80 end-users. The IT expert group consisted of 20 qualified respondents with relevant IT knowledge and experience.

4.2 Evaluation Scale

The researchers used a five-point Likert scale to evaluate the DC SPACE system. The scale was used to measure the level of agreement of the respondents with each statement in the questionnaire. The mean scores were interpreted using the corresponding mean range and descriptive rating shown below.

Table 4.2 Evaluation Scale

Numerical Scale	Mean Range	Description
5	4.51 – 5.00	Strongly Agree
4	3.51 – 4.50	Agree
3	2.51 – 3.50	Neutral
2	1.51 – 2.50	Disagree
1	1.00 – 1.50	Strongly Disagree

The scale was used in interpreting the results of both the end-user and expert evaluations. A higher mean score indicates a higher level of agreement with the statement, while a lower mean score indicates a lower level of agreement.

4.3 Statistical Treatment of Data

The collected responses were encoded using the numerical values assigned to the five-point Likert scale. The mean and standard deviation were computed for each questionnaire item and summarized under the eight software quality characteristics. Cronbach's alpha was computed separately for each respondent group and characteristic. Correlation was used to examine the similarity in the pattern of characteristic-level evaluations between the two respondent groups.

4.3.1 Mean

The arithmetic mean was used to determine the average rating for each questionnaire item and software quality characteristic. It was computed by dividing the sum of the numerical responses by the number of respondents.

Mean = $\Sigma X / n$

Where: ΣX = sum of the responses; n = number of respondents.

4.3.2 Standard Deviation

Standard deviation was used to determine how spread out the responses were from the mean. A lower standard deviation indicates that the respondents' ratings were more closely grouped, while a higher standard deviation indicates greater variation in their ratings. The sample standard deviation was computed using the responses from the respondents.

$$s = \sqrt{[\Sigma(X - \bar{X})^2 / (n - 1)]}$$

Where: X = an individual response

$\bar{X}$ = mean of the responses

n = number of respondents

4.3.3 Correlation

Correlation was used to examine whether the end-user and IT expert evaluations followed a similar pattern across the eight software quality characteristics. For this analysis, the mean score of each characteristic from the end-user group was paired with the mean score of the corresponding characteristic from the IT expert group. Pearson's correlation coefficient was computed using these eight pairs of characteristic-level means.

The correlation analysis is descriptive and exploratory because the two respondent groups were independent and the comparison used eight aggregate characteristic-level pairs. The result should not be interpreted as a relationship between individual end-users and individual IT experts, nor as evidence of causation.

4.3.4 Cronbach's Alpha

Cronbach's alpha was used to assess the internal consistency of the questionnaire items under each software quality characteristic. Each characteristic consisted of five items. Alpha was calculated separately for the end-user and IT expert responses and separately for each characteristic.

$$\alpha = [k / (k - 1)] [1 - (\Sigma\sigma^2_i / \sigma^2\square)]$$

Where: α = Cronbach's alpha

k = number of items

σ^2_i = variance of each item

$\sigma^2\square$ = variance of the total score for the items

For this study, each reliability coefficient was based on five items within one characteristic. The coefficients are reported separately for the two respondent groups.

4.4 Presentation, Analysis, and Interpretation of the Evaluation Results

The results are presented according to the eight software quality characteristics. The mean scores indicate the overall level of agreement, while the standard deviations show the variation in respondents' ratings. The results of the end-users and IT experts are presented separately.

4.4.1 Functional Suitability

Functional Suitability evaluates one aspect of the software quality of DC SPACE. The questionnaire contained five statements under this characteristic. The following results summarize the ratings of the end-users and IT experts.

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End Users' Responses

Functionality	Mean	Remarks
Suitability. The system provides the necessary functions for browsing/registering for events, RFID attendance, certificates, feedback, and other event-related tasks.		Highly Acceptable
Correctness. The system accurately records my event attendance through RFID.		Highly Acceptable
Accuracy. The event details, registration, attendance status, and certificates shown in the system match what I expect from my actions.		Highly Acceptable
Functional Appropriateness. The system allows me to access my certificates after meeting the required attendance conditions.		Highly Acceptable
Compliance. The system provides the expected school event features and access based on my role (student, organizer, faculty, or admin).		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Functionality	Mean	Remarks
Completeness. The system correctly		Highly Acceptable

implements the specified functional requirements, including event browsing/registration, RFID attendance, certificates, feedback, and event management.		
Correctness. The RFID attendance module accurately captures and processes participant attendance.		Highly Acceptable
Correctness. The system correctly manages event registration, participation, attendance, and certificate records.		Highly Acceptable
Functional Appropriateness. The system appropriately integrates attendance data with certificate eligibility and generation.		Highly Acceptable
Functional Completeness. The system follows defined functional requirements for authentication, role-based access, and event status/reporting.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.2 Performance Efficiency

Performance Efficiency evaluates the system's performance in relation to the time and resources used while performing its intended functions.

End Users' Responses

Performance Efficiency	Mean	Remarks
Time Behavior. The system responds quickly when I use its features.		Highly Acceptable
Resource Utilization. Event information is displayed promptly when I access it.		Highly Acceptable
Resource Utilization. My RFID attendance is recorded and reflected promptly in the system.		Highly Acceptable
Resource Utilization. The system remains responsive when accessing different pages or features.		Highly Acceptable
Efficiency Impact. The DC Space app makes event attendance, feedback, and certificate-related tasks easier and faster than doing them manually.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Performance Efficiency	Mean	Remarks
Time Behavior. The system responds within an acceptable amount of time when performing its functions.		Highly Acceptable
Resource Utilization. The system efficiently processes event and participant-related data.		Highly Acceptable

Time Behavior. RFID attendance data are processed and reflected within an acceptable response time.		Highly Acceptable
Resource Utilization. The system maintains acceptable performance when accessing multiple features.		Highly Acceptable
Efficiency Impact. The system reduces manual effort compared with traditional paper-based event attendance and certificate processes.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.3 Compatibility

Compatibility evaluates the system's ability to operate and interact appropriately within its intended environment.

End Users' Responses

Compatibility	Mean	Remarks
Interoperability. The system operates appropriately across its supported devices and platforms.		Highly Acceptable
Interoperability. The system's modules and components interact correctly with one another.		Highly Acceptable
Interoperability. Event registration, attendance,		Highly Acceptable

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certificates, and feedback functions are properly integrated.		
Data Integrity. The system maintains data consistency when information is transferred between connected modules.		Highly Acceptable
Co-existence. The system can operate alongside its required supporting services without significant conflicts.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Compatibility	Mean	Remarks
Interoperability. The system operates appropriately across its supported devices and platforms.		Highly Acceptable
Interoperability. The system's modules and components interact correctly with one another.		Highly Acceptable
Interoperability. Event registration, attendance, certificates, and feedback functions are properly integrated.		Highly Acceptable
Data Integrity. The		Highly Acceptable

system maintains data consistency when information is transferred between connected modules.		
Co-existence. The system can operate alongside its required supporting services without significant conflicts.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.4 Usability

Usability evaluates the extent to which users can effectively, efficiently, and satisfactorily interact with DC SPACE.

End Users' Responses

Usability	Mean	Remarks
Understandability. The system screens, buttons, labels, and instructions are easy to understand.		Highly Acceptable
Operability. It is easy to find the features and information I need.		Highly Acceptable
Recognizability. The system provides clear and understandable		Highly Acceptable

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feedback messages when an error occurs.		
Learnability. The system provides enough cues and feedback, such as confirmations, status labels, and navigation hierarchy, for new users to learn quickly.		Highly Acceptable
User Interface Aesthetics. The system looks organized and consistent across pages, which makes it easier to use.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Usability	Mean	Remarks
Appropriateness Recognizability. The user interface follows appropriate usability principles and uses clear layout, labels, icons, and navigation.		Highly Acceptable
Operability. Navigation elements are logically structured and consistent.		Highly Acceptable
Recognizability. System messages provide clear guidance when users encounter errors.		Highly Acceptable

Learnability. The system minimizes the learning effort required for new users.		Highly Acceptable
Operability. The interface design promotes efficient completion of user tasks and remains organized and consistent across modules.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.5 Reliability

Reliability evaluates the system's ability to perform its intended functions consistently under specified conditions.

End Users' Responses

Reliability	Mean	Remarks
Maturity. The system's core features operate without frequent unexpected failures.		Highly Acceptable
Fault Tolerance. The system prevents my entered information from		Highly Acceptable

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being lost if I make a mistake or click the wrong option.		
Recoverability. The system can restore a usable state after an interrupted session, page refresh, or temporary failure.		Highly Acceptable
Stability. The system remains stable during extended use without progressive slowdowns or repeated errors.		Highly Acceptable
Availability. The system is accessible when I need to use it.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Reliability	Mean	Remarks
Maturity. The system operates consistently without unexpected failures, freezing, or crashing during normal use.		Highly Acceptable
Fault Tolerance. The system handles invalid inputs and user errors without corrupting or losing data.		Highly Acceptable
Recoverability. The system provides		Highly Acceptable

appropriate recovery mechanisms when failures, interruptions, page refreshes, or temporary disconnections occur.		
Stability. The system maintains stable functionality during prolonged operation without progressive performance degradation.		Highly Acceptable
Availability. The system maintains data consistency during normal operations and remains accessible when needed.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.6 Security

Security evaluates the system's ability to protect information and data and control access according to authorized permissions.

End Users' Responses

Security	Mean	Remarks
Authenticity. The system secures my account access using my SDCA Gmail account and OTP verification.		Highly Acceptable

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Confidentiality. My personal information is protected by the system.		Highly Acceptable
Access Control. I can access only the features appropriate to my account role.		Highly Acceptable
Integrity. I am confident that my attendance records cannot be changed without proper authorization.		Highly Acceptable
Accountability. I am confident that important system actions are properly recorded.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Security	Mean	Remarks
Authenticity. The system provides a secure login process using the SDCA Gmail account and One-Time Password (OTP) verification.		Highly Acceptable
Confidentiality. The system adequately protects SDCA students', faculty members', and participants' personal information.		Highly Acceptable

Access Control. Access permissions are enforced according to user roles.		Highly Acceptable
Integrity. The system protects attendance and administrative records from unauthorized modification.		Highly Acceptable
Accountability. Administrative actions are logged and traceable to responsible user		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.7 Maintainability

Maintainability evaluates the extent to which DC SPACE can be modified, corrected, or improved efficiently when changes or issues arise.

End Users' Responses

Maintainability	Mean	Remarks
Modularity. System improvements and updates are implemented effectively.		Highly Acceptable
Modifiability. System updates do not significantly disrupt my use of the system, and the main features continue to work properly afterward.		Highly Acceptable
Analysability. Problems		Highly Acceptable

encountered while using the system are resolved appropriately.		
Adaptability. New features can be added without negatively affecting existing functions.		Highly Acceptable
Testability. The system remains usable during updates and maintenance activities.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Maintainability	Mean	Remarks
Modularity. The system architecture and components are organized to facilitate maintenance and future improvements.		Highly Acceptable
Modifiability. The source code, configuration, and system components are organized for maintainability and can be modified with limited disruption.		Highly Acceptable
Analysability. System faults and validation issues can be identified and corrected with reasonable effort through clear error messages, logs, and observable system behavior.		Highly Acceptable
Adaptability. System components are		Highly Acceptable

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sufficiently modular to support future enhancements without negatively affecting existing functions. .		
Testability. The system can be maintained and updated without compromising existing functionality.		Highly Acceptable
Overall Mean		Highly Acceptable

4.4.8 Portability

Portability evaluates the system's ability to operate effectively in different intended environments.

End Users' Responses

Portability	Mean	Remarks
Installability. I can access the system using different supported devices.		Highly Acceptable
Adaptability. The system works properly when accessed through different supported web browsers		Highly Acceptable
Adaptability. The system's features remain accessible when I switch between supported devices.		Highly Acceptable
Replaceability. The system maintains a consistent interface across supported devices and can be used without needing to install additional		Highly Acceptable

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software.		
Adaptability. My data remains available when I switch devices.		Highly Acceptable
Overall Mean		Highly Acceptable

Experts' Responses

Portability	Mean	Remarks
Installability. The system can be deployed across its supported platforms with minimal modification.		Highly Acceptable
Adaptability. The system can be installed and configured in its supported environments with reasonable effort.		Highly Acceptable
Adaptability. The system operates consistently across supported environments.		Highly Acceptable
Replaceability. Platform-specific dependencies are appropriately minimized, and the system can be migrated or redeployed to another supported environment with reasonable effort.		Highly Acceptable
Adaptability. The system can be migrated to another supported environment with minimal configurations.		Highly Acceptable
Overall Mean		Highly Acceptable

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4.5 Overall Evaluation of DC SPACE

The overall evaluation summarizes the results obtained from the end-users and IT experts across the identified software quality characteristics.

Software Quality Characteristics	End-User Mean	Expert Mean
Functional Suitability		
Performance Efficiency		
Compatibility		
Usability		
Reliability		
Security		
Maintainability		
Portability		
Overall Mean		

4.5.1 Correlation Between End-User and Expert Evaluation

Statistic	Result
Pearson correlation coefficient (r)	
Number of paired characteristics (n)	
p-value	

4.6 Reliability Testing of the Evaluation Questionnaire

Cronbach's alpha was computed separately for the end-user and IT expert questionnaires. The calculation was performed for each of the eight software quality characteristics using the five statements assigned to that characteristic.

Software Quality Characteristics	**End-User Alpha**	**Expert Alpha**
Functional Suitability		
Performance Efficiency		
Compatibility		
Usability		
Reliability		
Security		
Maintainability		
Portability		

CHAPTER 5

BIBLIOGRAPHY

Ahmad, N., Dzulkifli, D. Z. M., Raj, V. G., & Ishak, M. I. (2021). *Journal of telecommunication, electronic and computer engineering*. Universiti Teknikal Malaysia Melaka. https://jtmt.utem.edu.my/jtmt/article/view/6112

Aldabagh, M. (2024). *A review of students attendance management systems*. *Eurasian Journal of Science and Engineering, 10*(3), 136–145. https://eajse.tiu.edu.iq/index.php/eajse/article/download/455/405

Alshamaila, Y., Papagiannidis, S., & Li, F. (2020). *Cloud computing adoption by SMEs in the north east of England*. https://doi.org/10.1016/j.jbusres.2019.09.060

Azhari, A., & Supriyanto, S. (2023). *The importance of increasing attendance efficiency and accuracy with presence systems in the era of industrial revolution 4.0*. *International Journal of Computer and Information Technology Systems Management*. https://iiast.iaic-publisher.org/ijcitsm/index.php/IJCITSM/article/view/168/98

Bragdon, A., et al. (2020). *Touchscreen interaction design for public systems*. https://dl.acm.org/doi/10.1145/3313831.3376724

Bugingo, E., et al. (2025). *Enhancing face recognition attendance systems utilizing advanced techniques*. *Results in Engineering*. https://www.sciencedirect.com/science/article/abs/pii/S254266052500174X

Cornelio, B. L., Jr., Borcelo, R. S., & Siason, N. D., Jr. (2023). *Multiple events and multiple users tabulation system with e-certificate generation*. *International Journal of Advanced Science and Computer Engineering*. https://ijasce.org/index.php/IJASCE/article/view/240

Coronel, C., & Morris, S. (2020). *Database systems: Design, implementation, & management* (13th ed.). https://www.cengage.com/c/database-systems-design-implementation-management-13e-coronel/

Elmasri, R., & Navathe, S. (2021). *Fundamentals of database systems* (7th ed.). https://www.pearson.com/en-us/subject-catalog/p/fundamentals-of-database-systems/P200000003468

Goffar, E. A., & Arifah, R. P. N. (2023). *Certificate digitalization for information security in verification*. *Polytechnic Astra Proceedings*. https://proceedings.polytechnic.astra.ac.id/index.php/firstjournal/article/view/460

Godase, V., et al. (2025). *A review on real-time RFID-based smart attendance systems for efficient record management*. ResearchGate. https://www.researchgate.net/profile/Vaibhav-Godase/publication/394736438_A_Review_on_Real-Time_RFID-Based_Smart_Attendance_Systems_for_Efficient_Record_Management/links/68a59afe2c7d3e0029b21b3d/A-Review-on-Real-Time-RFID-Based-Smart-Attendance-Systems-for-Efficient-Record-Management.pdf

Gulliksen, J., Göransson, B., Boivie, I., Persson, J., Blomkvist, S., & Cajander, Å. (2021). *Key principles for user-centered systems design*. https://www.sciencedirect.com/science/article/pii/S1071581920301015

Huliman, Wijaya, R., & Ayu. (2024). *Design of a website-based student e-certificate system using the user centered design (UCD) method*. *Journal of Artificial Intelligence and Educational Applications*. https://mail.ioinformatic.org/index.php/JAIEA/article/view/1551

Ishaq, K., & Bibi, S. (2023). *IoT-based smart attendance system using RFID: A systematic literature review*. arXiv. https://arxiv.org/abs/2308.02591

Johan, M. C. (2020). *E-certificate system based on portable document format and QR code for academic activities*. *International Journal of Computer Science Issues*. https://d1wqtxts1xzle7.cloudfront.net/115569588/IJCSI-17-6-87-91-libre.pdf

Kamis, T., Md Nor, A. R., Adam, A., & Ismail, M. S. (2024). *Electronic certificate acceptance based on technology acceptance model (TAM)*. *Ecosia Conference Proceedings*. https://conference.asia.ac.id/index.php/ecosia/article/view/185

Khan, A. N., Khan, N. A., & Soomro, M. A. (2021). *Cloud computing adoption in higher education institutions*. https://doi.org/10.1016/j.tele.2020.101592

Kumar, R., et al. (2025). *A modern web-based student attendance management system*. *International Journal of Innovative Research in Technology, 11*(10). https://ijirt.org/publishedpaper/IJIRT174142_PAPER.pdf

Lazar, J., Feng, J., & Hochheiser, H. (2021). *Research methods in human-computer interaction*. https://www.sciencedirect.com/book/9780128053904/research-methods-in-human-computer-interaction

Mitigating attendance fraud through multi-biometric authentication: A comparative study of fingerprint and facial recognition technologies. (2024). *Scientific Journal of Applied Sciences and Research*. http://ssaapublications.com/index.php/sjasor/article/view/394/363

Mondal, S. (2023). *Digital certification and smart verification systems*. *Science and Culture*. https://scienceandculture-isna.org/uploads/latestIssue/07_Soumen_Mondal.pdf

Muhtar, A. N. A. (2024). *E-certificates in schools to improve administrative efficiency in the Society 5.0 era*. *Management of Islamic Education Journal*. https://e-journal.staima-alhikam.ac.id/mpi/article/view/4089/1304

Naen, M. F., Adnan, M. H. M., Yazi, N. A., & Nee, C. K. (2023). *Development of attendance monitoring system with artificial intelligence optimization in cloud*. *International Journal of Artificial Intelligence*. https://www.lamintang.org/journal/index.php/ijai/article/view/315/224

Nadhan, A. S., Tukkoji, C., Shyamala, B., Lal, N. D., Kumar, A. N. S., Gowda, V. M., Adhoni, Z. A., & Endaweke, M. (2022). *Smart attendance system using IoT and artificial intelligence*. *Journal of Sensors*. https://onlinelibrary.wiley.com/doi/10.1155/2022/4899768

Nithya, J., Vignesh, S. R., Devadarsan, A., & Venkatesh, S. (2023). *AI-based contactless attendance monitoring system*. Neliti. https://media.neliti.com/media/publications/431063-ai-based-contactless-attendance-monitori-0dbd2220.pdf

Qureshi, M. R. J. (2020). *Smart attendance system research paper*. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3635316

Reddy, N. K. P., et al. (2020). *AI-based attendance monitoring system*. ResearchGate. https://www.researchgate.net/profile/Naveen-Reddy-K-P-2/publication/338331249_AI-Based_Attendance_Monitoring_System/links/5e0ce58792851c8364ab7dda/AI-Based-Attendance-Monitoring-System.pdf

Sharma, P., & Kaur, G. (2022). *Cloud-based IoT framework for smart attendance systems*. https://doi.org/10.1016/j.iot.2021.100498

Sommerville, I. (2020). *Software engineering* (10th ed.). https://www.pearson.com/en-us/subject-catalog/p/software-engineering/P200000003480

Suwita, F. S. (2024). *Electronic certificate information system as media for distributing certificates online during the COVID-19 pandemic*. *Injuratech Journal*. https://ojs.unikom.ac.id/index.php/injuratech/article/view/14197

Thai, H.-D., Seo, Y.-S., & Huh, J.-H. (2024). *RFID and AI-based attendance system*. IEEE Xplore. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10763516

Vadwala, A. (2024). *A comprehensive review of digitalized attendance systems for various institutions*. *International Journal of Engineering Applied Sciences and Technology, 9*(3), 119–123.

Venkatesh, V., Thong, J. Y., & Xu, X. (2021). *Unified theory of acceptance and use of technology (UTAUT2)*. https://doi.org/10.1016/j.jmis.2020.11.002

Wiley. (2022). *AI-based contactless attendance monitoring system*. *Journal of Sensors*. https://onlinelibrary.wiley.com/doi/10.1155/2022/4899768

Winarti, T. (2023). *Web-based document certification system with advanced encryption standard digital signature*. https://d1wqtxts1xzle7.cloudfront.net/104662406/14852-libre.pdf

Al-Sharafi, M. A., AlAjmi, Q., Al-Emran, M., Qasem, Y. A. M., & Aldheleai, Y. M. (2021). Cloud computing adoption in higher education: An integrated theoretical model. In Recent Advances in Technology Acceptance Models and Theories (pp. 191–209). Springer.

Aydin, H. (2021). A study of cloud computing adoption in universities as a guideline to cloud migration. SAGE Open, 11(3). https://doi.org/10.1177/21582440211030280

Dayana, K. R., & Rani, P. S. (2023). Secure cloud data storage solution with better data accessibility and time efficiency. Automatika, 64(4), 756–763. https://doi.org/10.1080/00051144.2023.2213564

Naen, M. F., Adnan, M. H. M., Yazi, N. A., & Nee, C. K. (2021). Development of attendance monitoring system with artificial intelligence optimization in cloud. International Journal of Artificial Intelligence, 8(2), 88–98. https://doi.org/10.36079/lamintang.ijai-0802.315

Bhardwaj, A. K., Garg, L., Garg, A., & Gajpal, Y. (2021). E-learning during COVID-19 outbreak: Cloud computing adoption in Indian public universities. Computers, Materials & Continua, 66(3), 2471–2492. https://doi.org/10.32604/cmc.2021.014099

Taipalus, T. (2024). Database management system performance comparisons: A systematic literature review. Journal of Systems and Software, 208, Article 111872. https://doi.org/10.1016/j.jss.2023.111872

Jose, B., & Abraham, S. (2020). Performance analysis of NoSQL and relational databases with MongoDB and MySQL. Materials Today: Proceedings, 24, 2036–2043. https://doi.org/10.1016/j.matpr.2020.03.634

Barrocan, R. A., Calzo, M. J. R., Carambas, C. B., Tugade, M. J. E., & Bartolome, M. B. (2025). An architectural approach of a web-based monitoring system for efficient internship time tracking. International Journal of Research and Innovation in Applied Science, 10(2), 285–289. https://doi.org/10.51584/IJRIAS.2025.10020023

Popa, V., Petrariu, A. I., Mutescu, P. M., Maftei, A. A., & Lavric, A. (2026). Standalone RFID access control system with data-integrity verification capabilities. Sensors, 26(9), Article 2892. https://doi.org/10.3390/s26092892

Gulliksen, J., Göransson, B., Boivie, I., Blomkvist, S., Persson, J., & Cajander, Å. (2003, corrected from 2021). Key principles for user-centered systems design. Behaviour & Information Technology, 22(6), 397–409. https://doi.org/10.1080/01449290310001624329

Xue, L., Rashid, A. M., & Ouyang, S. (2024). The Unified Theory of Acceptance and Use of Technology (UTAUT) in higher education: A systematic review. *SAGE Open, 14*. https://doi.org/10.1177/21582440241229570

Kamis, T., Md Nor, A. R., Adam, A., & Ismail, M. S. (2025, corrected from 2024). Electronic certificate acceptance based on technology acceptance model (TAM). Proceeding Economy of Asia International Conference, 2025(1). https://conference.asia.ac.id/index.php/ecosia/article/view/185

Abdul Halim, M. A., Zamzuri, M. A. M., & Ghazali, N. A. (2023). The transformative role of artificial intelligence in the event management industry. Journal of Islamic, Social, Economics and Development. https://journal.uitm.edu.my/ojs/index.php/JIBE/article/view/1015

Alasmari, S., Farooqi, A. A., & Alotaibi, M. (2024). Recent trends in crowd management using deep learning techniques: A systematic literature review. Journal of Computer Science and Technology. https://link.springer.com/article/10.1007/s43995-024-00071-3

Bhuiyan, E. A., et al. (2021). Crowd density estimation using deep learning for Hajj pilgrimage video analytics. F1000Research, 10, 1190. https://f1000research.com/articles/10-1190/v1

Camacho-Ruiz, A., et al. (2023). Tourism destination events classifier based on artificial intelligence techniques. Applied Soft Computing. https://www.sciencedirect.com/science/article/pii/S1568494623009328

Causin, G. F., & Scamacca, A. (2021). Technology in the meetings and events industry. In M3: The Future of Meetings and Events. https://digitalcommons.usf.edu/m3publishing/vol17/iss9781732127593/1/

Khan, M. A., Menouar, H., & Hamila, R. (2023). LCDnet: A lightweight crowd density estimation model for real-time video surveillance. Journal of Real-Time Image Processing. https://link.springer.com/article/10.1007/s11554-023-01286-8

Ryan, R., et al. (2020). Recognizing Events 4.0: The digital maturity of events. Event Management. https://www.stir.ac.uk/research/hub/publication/1759238

Xiang, X. (2022). Crowd density estimation method using deep learning for passenger flow detection system in exhibition center. Computational Intelligence and Neuroscience. https://onlinelibrary.wiley.com/doi/10.1155/2022/1990951

Yamin, M., et al. (2023). Sparrow search optimization with transfer learning-based crowd density classification. Computers, Materials & Continua, 74(3). https://www.techscience.com/cmc/v74n3/50924

Zamzuri, M. A. M., et al. (2023). Conceptualising event technology usage and experience in event management: Thematic analysis. Asia-Pacific Journal of Innovation in Hospitality and Tourism. https://fslmjournals.taylors.edu.my/wp-content/uploads/APJIHT/APJIHT-2023-12-3-Special-Issue/APJIHT-123_full-issue.pdf

Brescia-Zapata, M. (2023). Towards sustainable and accessible events: An exploratory research perspective. Universal Access in the Information Society. https://link.springer.com/article/10.1007/s10209-023-01069-9

Chen, C., & Jung, T. (2024). Event technology and attendee experience: A systematic review and research agenda. https://via.library.depaul.edu/ichrie_rr/vol9/iss1/4/

Chen, C., Jung, T., & Cai, W. (2025). A critical review of technology-facilitated event engagement: Current landscape and pathway forward. https://doi.org/10.1108/IJCHM-02-2025-0236

Neuhofer, B., Magnus, A., & Celuch, K. (2021). The impact of artificial intelligence on event experiences: A scenario technique approach. https://doi.org/10.1007/s12525-020-00433-4

Ramli, N. A., et al. (2023). Factors influencing technology usage among event attendees. International Journal of Business and Society. https://ojs.amhinternational.com/index.php/imbr/article/view/3465

Selman, L. E., et al. (2023). Engaging and supporting the public on the topic of grief and bereavement: An evaluation of Good Grief Festival. https://journals.sagepub.com/doi/abs/10.1177/26323524231189523

Zamzuri, M. A. M., et al. (2023). Conceptualising event technology usage and experience in event management: Thematic analysis. Asia-Pacific Journal of Innovation in Hospitality and Tourism. https://fslmjournals.taylors.edu.my/wp-content/uploads/APJIHT/APJIHT-2023-12-3-Special-Issue/APJIHT-123_full-issue.pdf